

FC7xxx SPI User Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.3	2023/10/20	Updated for MCAL V0.3.0
0.4	2023/11/22	Updated for MCAL V0.4.0
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Chapter 1 SPI Introduction

1.1 Requirements

The design of this module follows the specifications of the SPI driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_SWS_SPIHandlerDriver*.

1.2 Hardware Summary

The SPI Handler and Driver provide services for reading from and writing to devices connected via SPI buses. It provides access to SPI communication to several users. It also provides the required mechanism to configure the on-chip SPI peripheral.

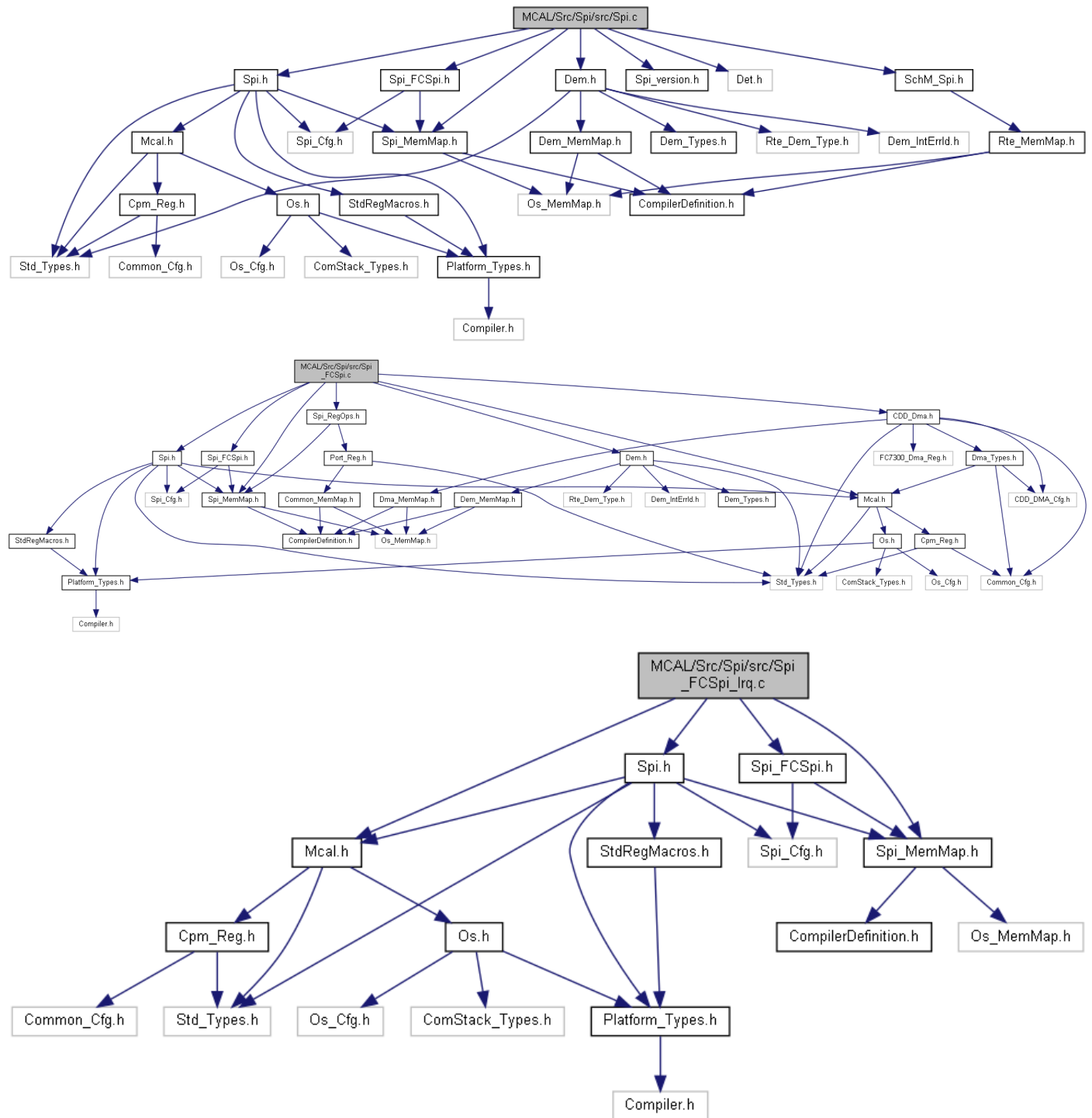
This specification describes the API, Mapping to SWS requirements for a monolithic SPI Handler and Driver. This software module includes handling and driving functionalities. Main objectives of this monolithic SPI Driver are to take the best of each microcontroller features and to allow implementation optimization depending on static configuration to fit as much as possible to ECU needs.

The general behavior of the SPI Handler and Driver could be asynchronous or synchronous according to the level of functionality selected.

The specification covers the Handler and Driver functionalities combined in one single module. The SPI handler controls multiple accesses to buses that could be located in the ECU Abstraction layer. The other part is the SPI driver that accesses the microcontroller hardware directly that could be located in the Microcontroller Abstraction layer.

Chapter 2 Software Design

2.1 File Structure



2.2 Macros

2.2.1 Macros in Spi.h

- `#define SPI_PHYUNIT_SYNC_U8 ((uint8)1)`
Indicates the HW unit is configured as Sync mode.

- **#define SPI_PHYUNIT_ASYNC_U8** ((uint8)0)
Indicates the HW unit is configured as Async mode.
- **#define SPI_E_PARAM_CHANNEL** ((uint8)0x0Au)
Function called with wrong parameter of channel.
- **#define SPI_E_PARAM_JOB** ((uint8)0x0Bu)
Function called with wrong parameter of job.
- **#define SPI_E_PARAM_SEQ** ((uint8)0x0Cu)
Function called with wrong parameter of sequence.
- **#define SPI_E_PARAM_LENGTH** ((uint8)0x0Du)
Function called with wrong parameter of external buffer.
- **#define SPI_E_PARAM_CONFIG** ((uint8)0x0Fu)
API service called with wrong resource assigned
- **#define SPI_E_PARAM_UNIT** ((uint8)0x0Eu)
Function called with wrong parameter of HW unit.
- **#define SPI_E_UNINIT** ((uint8)0x1Au)
Function called without module initialization.
- **#define SPI_E_SEQ_PENDING** ((uint8)0x2Au)
Indicates sequence status is pending or used HW unit is busy.
- **#define SPI_E_SEQ_IN_PROCESS** ((uint8)0x3Au)
Synchronous transmission service called at wrong time: HW unit is already running.
- **#define SPI_E_ALREADY_INITIALIZED** ((uint8)0x4Au)
Function called while the SPI driver has already been initialized.
- **#define SPI_E_CONFIG_OUT_OF_RANGE** ((uint8)0x5Au)
The number of sequences, jobs or channels exceeds precompile time sizes.
- **#define SPI_E_PARAM_EB_UNIT** ((uint8)0x5Bu)
Sequence contains uninitialized external buffers.

- **#define SPI_E_SEQ_EMPTY** ((uint8)0x5Cu)
Sequence is empty.
- **#define SPI_E_JOB_EMPTY** ((uint8)0x5Du)
Job is empty.
- **#define SPI_E_PARAM_POINTER** ((uint8)0x10u)
The parameter version info is NULL_PTR.
- **#define SPI_INIT_ID** ((uint8) 0x00u)
SPI Init function ID.
- **#define SPI_DEINIT_ID** ((uint8) 0x01u)
SPI Delnit function ID.
- **#define SPI_WRITEIB_ID** ((uint8)0x02u)
SPI writes IB function ID.
- **#define SPI_ASYNCTRANSMIT_ID** ((uint8)0x03u)
SPI async transmit function ID.
- **#define SPI_READIB_ID** ((uint8)0x04u)
SPI reads IB function ID.
- **#define SPI_GETSTATUS_ID** ((uint8)0x05u)
SPI gets status function ID.
- **SPI_GETSTATUS_ID** ((uint8) 0x06u)
API service ID for SPI get status function
- **#define SPI_GETJOBRESULT_ID** ((uint8)0x07u)
SPI gets Job result function ID.
- **#define SPI_GETSEQUENCERESULT_ID** ((uint8)0x08u)
SPI gets sequence result function ID.
- **#define SPI_GETVERSIONINFO_ID** ((uint8)0x09u)
SPI gets version info function ID.

- **#define SPI_SYNCTRANSMIT_ID** ((uint8) 0x0Au)
SPI syncs transmit function ID.
- **#define SPI_GETHWUNITSTATUS_ID** ((uint8) 0x0Bu)
SPI gets HW status function ID.
- **#define SPI_CANCEL_ID** ((uint8) 0x0Cu)
SPI cancels function ID.
- **#define SPI_SETASYNCMODE_ID** ((uint8) 0x0Du)
SPI sets async mode function ID.
- **#define SPI_MAINFUNCTION_HANDLING_ID** ((uint8)0x10u)
SPI main function ID.
- **#define SPI_SETHWUNITASYNCMODE_ID** ((uint8)0x80u)
SPI sets HW Unit async mode function ID.
- **#define SPI_JOB_PRIORITY_LEVELS_COUNT** 4
The number of allowed job priority levels (0..3).

2.2.2 Macros in Spi_version.h

- **#define SPI_AR_RELEASE_MAJOR_VERSION** 4
- **#define SPI_AR_RELEASE_MINOR_VERSION** 6
- **#define SPI_AR_RELEASE_REVISION_VERSION** 0
- **#define SPI_SW_MAJOR_VERSION** 0
- **#define SPI_SW_MINOR_VERSION** 6
- **#define SPI_SW_PATCH_VERSION** 0
- **#define SPI_VENDOR_ID** 174
- **#define SPI_MODULE_ID** 83

2.2.3 Macros in Spi_Cfg.h

- #define **SPI_CFG_AR_RELEASE_MAJOR_VERSION** 4

- #define **SPI_CFG_AR_RELEASE_MINOR_VERSION** 6

- #define **SPI_CFG_AR_RELEASE_REVISION_VERSION** 0

- #define **SPI_CFG_MODULE_ID** 83

- #define **SPI_CFG_SW_MAJOR_VERSION** 0

- #define **SPI_CFG_SW_MINOR_VERSION** 6

- #define **SPI_CFG_SW_PATCH_VERSION** 0

- #define **SPI_CFG_VENDOR_ID** 174

- #define **SPI_MAX_PARTITIONS** (1U)

- #define **SPI_MULTICORE_ENABLED** (STD_OFF)

- #define **SPI_SPURIOUS_CORE_ID_U32** ((uint32)0UL)

- #define **SPI_DET_ENABLE** (STD_OFF)
Switches the Development Error functionality ON or OFF.

- #define **SPI_GET_VERSION_INFO_API_SUPPORT** (STD_OFF)
Switches the Version Information API functionality ON or OFF.

- #define **SPI_INTERRUPTIBLE_SEQ_ALLOWED** (STD_OFF)
Switches the Interruptible Sequences handling functionality ON or OFF.

- #define **SPI_CANCEL_API** (STD_ON)
Switches the Spi_Cancel function ON or OFF.

- #define **SPI_HW_STATUS_API** (STD_ON)
Switches the Spi_GetHWUnitStatus function ON or OFF.

- **#define SPI_CHANNEL_BUFFER_TYPE_FLAG_ONLY_IB 0u**
Only the internal buffer is allowed for channel.
- **#define SPI_CHANNEL_BUFFER_TYPE_FLAG_ONLY_EB 1u**
Only the external buffer is allowed for channel.
- **#define SPI_CHANNEL_BUFFER_TYPE_FLAG_IB_EB_BOTH 2u**
the SPI Handler/Driver manages both buffers type
- **#define SPI_CHANNEL_BUFFERS_ALLOWED (SPI_CHANNEL_BUFFER_TYPE_FLAG_ONLY_IB)**
Selects the SPI Handler Driver Channel Buffers usage allowed and delivered.
- **#define SPI_DELIVER_LEVEL_0 0x00u**
The LEVEL 0 Simple Synchronous SPI Handler Driver functionalities are selected.
- **#define SPI_DELIVER_LEVEL_1 0x01u**
The LEVEL 1 Basic Asynchronous SPI Handler Driver functionalities are selected.
- **#define SPI_DELIVER_LEVEL_2 0x02u**
The LEVEL 2 Enhanced SPI Handler Driver functionalities are selected.
- **#define SPI_LEVEL_DELIVERED (SPI_DELIVER_LEVEL_2)**
Selects the SPI Handler Driver level of scalable functionality.
- **#define SPI_ASYNC_DRIVER (STD_ON)**
Support asynchronous driver or not
- **#define SPI_SYNC_DRIVER (STD_ON)**
Support synchronous driver or not
- **#define SPI_CHANNEL_MAX_COUNT (4u)**
Defines the maximum number of supported channels.
- **#define SPI_JOB_MAX_COUNT (3u)**
Total number of Jobs configured.
- **#define SPI_SEQUENCE_MAX_COUNT (3u)**
Total number of Sequences configured.

- **#define CSIB0** ((uint8)0u)
 - **#define CSIB1** ((uint8)1u)
 - **#define CSIB2** ((uint8)2u)
- Defines the peripherals used throughout the configuration(s).
-
- **#define SPI_MODE_SYNC** 0u
- Define the operation mode
-
- **#define SPI_MODE_ASYNC_FIFO** 1u
- Define the operation mode
-
- **#define SPI_MODE_ASYNC_DMA** 2u
- Define the operation mode
-
- **#define SPI_HW_MODULE_MAX_COUNT** (1u)
- Defines the peripherals used throughout the configuration(s).
-
- **#define FCSPI_1_HWUNIT** (CSIB0)
 - **#define FCSPI_2_HWUNIT** (CSIB1)
 - **#define FCSPI_5_HWUNIT** (CSIB2)
- Physical HWUnits definition.
-
- **#define FCSPI_0_ENABLED** (STD_OFF)
 - **#define FCSPI_1_ENABLED** (STD_ON)
 - **#define FCSPI_2_ENABLED** (STD_ON)
 - **#define FCSPI_3_ENABLED** (STD_OFF)
 - **#define FCSPI_4_ENABLED** (STD_OFF)
 - **#define FCSPI_5_ENABLED** (STD_ON)
 - **#define FCSPI_6_ENABLED** (STD_OFF)
 - **#define FCSPI_7_ENABLED** (STD_OFF)
- Physical HWUnits enabled/disabled.
-
- **#define SPI_SLAVE_MODE_SUPPORT** (STD_ON)
- Physical HWUnit is configured to slave mode.
-
- **#define FCSPI_1_OFFSET** 1u
 - **#define FCSPI_2_OFFSET** 2u
 - **#define FCSPI_5_OFFSET** 5u
-
- **#define SPI_DEM_ENABLE** (STD_OFF)
- Switches the Production Error Detection and Notification OFF.

- **#define SPI_SUP_CONT_EN_SECOND_EDGE_SAMPLE (STD_ON)**
Define whether this chip support continuous mode when sampling on the second edge

- **#define SPI_VARIANT_PRECOMPILE (0)**
#define SPI_VARIANT_POSTBUILD (1)
#define SPI_VARIANT_LINKTIME (2)
Define values for Autosar configuration variants.

- **#define SPI_CONFIG_VARIANT (SPI_VARIANT_POSTBUILD)**
Defines the use of Pre-Compile(PC) support.

- **#define SPI_PRECOMPILE_SUPPORT (STD_OFF)**
Defines precompile support if VariantPreCompile or VariantLinkTime is selected and the number of variant <=1.

- **#define SPI_DMA_USED (STD_ON)**
Defines if transfers are made using DMA or FIFO.

- **#define SPI_WAIT_LOOP_TICKS 23u**
Number of CPU clock cycles consumed by a wait loop during the wait for TX/RX transmission to complete one frame in Spi_SyncTransmit().

- **#define SPI_TIMEOUT_COUNTER ((sint32)((12000000u) / SPI_WAIT_LOOP_TICKS))**
Defines the "Number of Loops" timeout.

- **#define SPI_SUPPORT_CONCURRENT_SYNC_TRANSMIT (STD_OFF)**
Allows simultaneous calls to Spi_SyncTransmit() for different threads.

- **#define SPI_HWUNIT_ASYNC_MODE (STD_ON)**
Allows or forbids simultaneous calls to Spi_SyncTransmit().

- **#define SPI_MORE_THAN_256_SEQ_JOB_CHAN_SUPPORT (STD_ON)**
If enabled, allows dual MCU clock configuration settings.

- **#define SPI_DATA_RW_8_BITS_PER_UNIT ((uint8)1)**
#define SPI_DATA_RW_16_BITS_PER_UNIT ((uint8)2)
#define SPI_DATA_RW_32_BITS_PER_UNIT ((uint8)4)
Defines the bit count of reading from/writing to memory per time.

2.3 Enums

2.3.1 Enumerations in Spi.h

2.3.1.1 Spi_JobResultType

Enumeration	Spi_JobResultType	
Description	A range of specific Jobs status for SPI Driver.	
Values	Value	Description
	SPI_JOB_OK = 0	The last transmission of the Job has been finished successfully.
	SPI_JOB_PENDING = 1	The SPI handler/Driver is performing a SPI Job.
	SPI_JOB_FAILED = 2	The last transmission of the Job has failed.
	SPI_JOB_QUEUED = 3	An asynchronous transmit Job has been accepted, while actual transmission for this Job has not started.

2.3.1.2 Spi_SeqResultType

Enumeration	Spi_SeqResultType	
Description	A range of specific Sequences status for SPI Driver.	
Values	Value	Description
	SPI_SEQ_OK = 0	The last transmission of the Sequence has been finished successfully
	SPI_SEQ_PENDING = 1	The SPI handler/Driver is performing an SPI Sequence.
	SPI_SEQ_FAILED = 2	The last transmission of the Sequence has failed.
	SPI_SEQ_CANCELLED = 3	The last transmission of the Sequence has been canceled by the user.

2.3.1.3 Spi_AsyncModeType

Enumeration	Spi_AsyncModeType	
Description	Specifies the asynchronous mode for SPI buses handled asynchronously in Level 2.	
Values	Value	Description
	SPI_POLLING_MODE = 0	The asynchronous mechanism is ensured by polling, so interrupts related to SPI buses handled asynchronously are disabled.
	SPI_INTERRUPT_MODE = 1	The asynchronous mechanism is ensured by interrupt, so interrupts related to SPI buses handled asynchronously are enabled

2.3.1.4 SPI_MasterSlaveModeType

Enumeration	SPI_MasterSlaveModeType	
Description	Specifies the operation mode (Master or Slave)	
Values	Value	Description
	SPI_MASTER = 0	SPI Hardware selected as MASTER
	SPI_SLAVE = 1	SPI Hardware selected as SLAVE

2.3.1.5 SPI_IbEbBufferType

Enumeration	SPI_IbEbBufferType	
Description	The enumeration containing the designated values for buffer types (internal or external).	
Values	Value	Description
	IB = 0	Internal buffer.
	EB = 1	External buffer.

2.3.1.6 Spi_StatusType

Enumeration	Spi_StatusType	
Description	A range of specific status for SPI Driver	
Values	Value	Description
	SPI_UNINIT = 0	Not initialized or not usable.
	SPI_IDLE = 1	Not currently transmitting any jobs.
	SPI_BUSY = 2	Is performing an SPI Job (transmit).

2.4 Typedefs

2.4.1 Typedefs in Spi.h

- **typedef uint8 Spi_DataBufferType**
Type of application data buffer elements.
- **typedef uint16 Spi_NumberOfDataType**
Type for defining the number of data elements of the type Spi_DataBufferType.
- **typedef uint32 Spi_ChannelType**
Specifies the identification (ID) for a Channel.
- **typedef uint32 Spi_JobType**
Specifies the identification (ID) for a Job.
- **typedef uint32 Spi_SequenceType**
Specifies the identification (ID) for a sequence of jobs.
- **typedef uint8 Spi_HWUnitType**
Specifies the identification(ID) for a SPI Hardware microcontroller peripheral (unit).
- **typedef uint8 Spi_ExternalDeviType**
Contains the ID of an external device.

- typedef **void (*Spi_PfnNotifyType) (void)**
Sequence end notification handler type.

2.5 Structures

2.5.1 Spi_ChnlAttrsType

Structure	Spi_ChnlAttrsType
Description	Internal structure used to manage the channel attributes.
Diagram	N/A
Data Fields	- uint32 SpiCoreUse CoreID used - uint32 u32ChnlTRCTRL Channel attributes for CMD in TCR register. - uint32 u32ChnlFIFOWTM Channel attributes for FIFO control in FCR register. - uint8 u8DataAccessed Channel attributes for data width.

2.5.2 SPI_ChannelsAttrsType

Structure	SPI_ChannelsAttrsType
Description	This structure holds the channel attributes configuration.
Diagram	<pre> graph BT SPI_ChannelsAttrsType[SPI_ChannelsAttrsType] -.-> ChannelAttrsCfg Spi_ChnlAttrsType[Spi_ChnlAttrsType] </pre>
Data Fields	const Spi_ChnlAttrsType * ChannelAttrsCfg Array of configuration channel attributes.

2.5.3 Spi_DeviceAttrType

Structure	Spi_DeviceAttrType
Description	Internal structure used to manage the FCSPI unit attributes.
Diagram	N/A
Data Fields	- uint32 SpiCoreUse CoreID used - uint32 u32TR_CTRL Device attributes for TR_CTRLs. - uint32 u32CLK_CFG Device configuration for timing in CCR. - uint32 u32CFGR1 Device CS idle setting

2.5.4 Spi_ExDevicesAttrsType

Structure	Spi_ExDevicesAttrsType
Description	This structure holds the External devices configuration.
Diagram	<pre> graph BT A[Spi_ExDevicesAttrsType] -.-> DeviceAttrsCfg B[Spi_DeviceAttrType] </pre>
Data Fields	const Spi_DeviceAttrType * DeviceAttrsCfg

2.5.5 Spi_SeqCfgType

Structure	Spi_SeqCfgType
Description	This structure contains all the needed data to configure one SPI Sequence.
Diagram	<pre> graph BT A[Spi_SeqCfgType] --> B[Spi_SequenceConfigType_struct] </pre>
Data Fields	- Spi_JobType tNumJobs Count of jobs in this sequence. - uint32 SpiCoreUse CoreID used - const Spi_JobType *pcJobsArray Job index list. - Spi_PfnNotifyType pfnSeqEndNotify Sequence end notification handler. - uint8 u8Interruptible Boolean indicating if the Sequence is interruptible or not.

2.5.6 Spi_SeqsCfgType

Structure	Spi_SeqsCfgType
Description	This structure holds the External devices configuration.

Diagram	<pre> graph TD A[Spi_SequenceConfigType_struct] --> B[Spi_SeqCfgType] B -.-> SequenceCfg C[Spi_SeqsCfgType] </pre>
Data Fields	const Spi_SeqCfgType * SequenceCfg

2.5.7 Spi_SeqStatType

Structure	Spi_SeqStatType
Description	Internal structure used to manage the sequence state.
Diagram	<pre> graph TD A[Spi_SequenceConfigType_struct] --> B[Spi_SeqCfgType] B -.-> pcSeqCfg C[Spi_SeqStateType_struct] C --> D[Spi_SeqStateType] </pre>
Data Fields	- Spi_SeqResultType eSeqResult Seq Result. - const Spi_SeqCfgType *pcSpiSeqCfg Pointer to the configuration. - const Spi_JobTyp *pcCurJobIdxPointer Position in JobIndexList to the job in transmission of an async sequence. - Spi_JobType tLeftJobs Number of jobs in a pending async sequence, not yet transmitted.

2.5.8 Spi_JobStatType

Structure	Spi_JobStatType
Description	Internal structure used to manage the job state.

Diagram	<pre> graph TD Spi_JobStateType[Spi_JobStateType] -.-> pAsyncCurSeqStat Spi_SeqStateType[Spi_SeqStateType] Spi_SeqStateType -.-> pcSeqCfg Spi_SeqCfgType[Spi_SeqCfgType] Spi_SeqCfgType --> Spi_SequenceConfigType_struct[Spi_SequenceConfigType_struct] </pre>
Data Fields	• Spi_JobResultType eJobResult Job Result • Spi_SeqStateType *pAsyncCurSeqStat state info of the async sequence that job belong to • Spi_JobType tAsyncNextWaitSchedJobIdx the next async job waiting to be scheduled to transmit

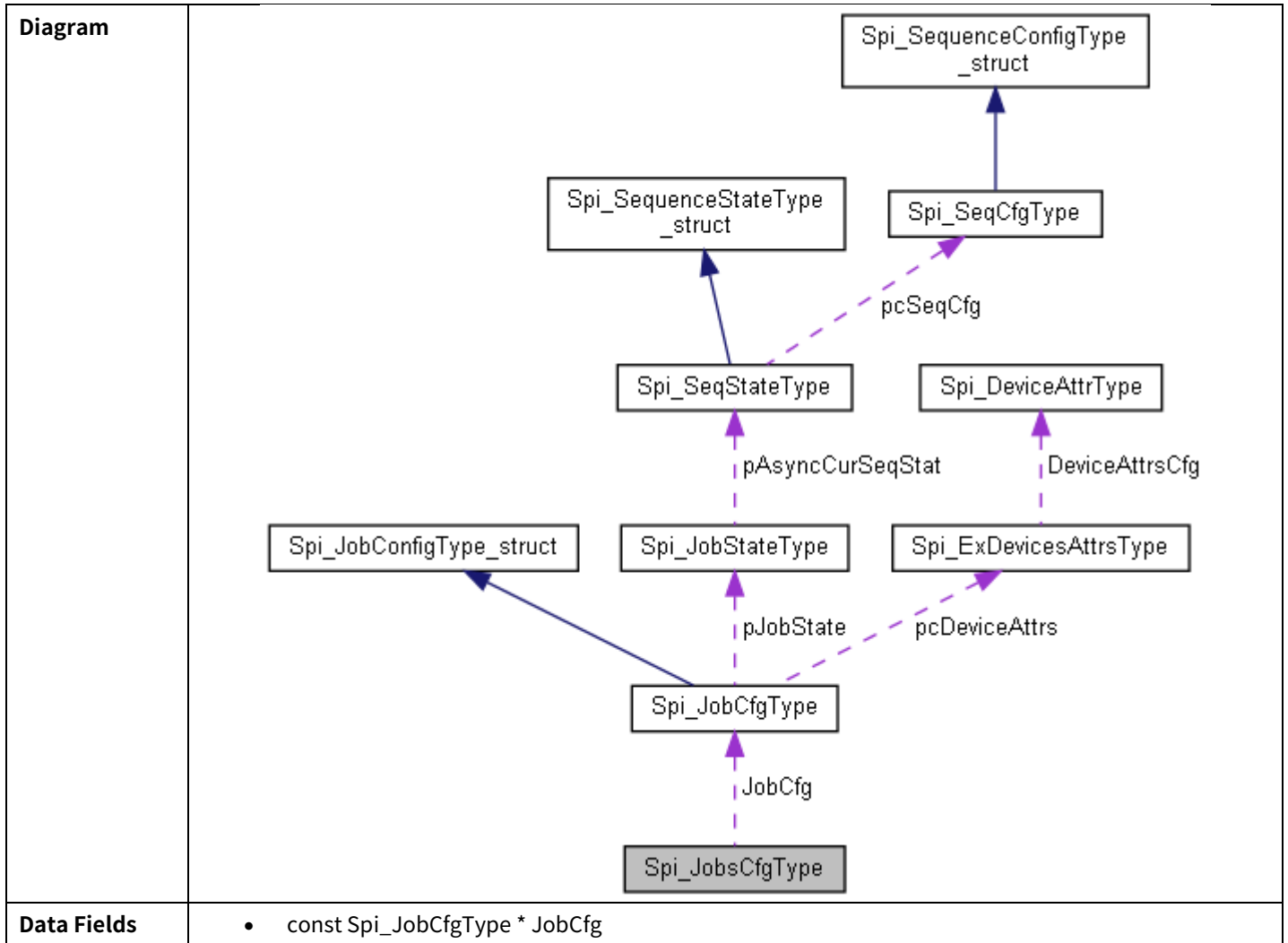
2.5.9 Spi_JobCfgType

Structure	Spi_JobCfgType
Description	The structure containing all parameters needed to completely define a Job.
Diagram	<pre> graph TD Spi_JobCfgType[Spi_JobCfgType] -.-> pJobState Spi_JobStateType[Spi_JobStateType] Spi_JobStateType -.-> pAsyncCurSeqStat Spi_SeqStateType[Spi_SeqStateType] Spi_SeqStateType -.-> pcSeqCfg Spi_SeqCfgType[Spi_SeqCfgType] Spi_SeqCfgType --> Spi_SequenceConfigType_struct[Spi_SequenceConfigType_struct] Spi_JobCfgType -.-> DeviceAttrsCfg Spi_DeviceAttrType[Spi_DeviceAttrType] Spi_JobCfgType -.-> pcDeviceAttrs Spi_ExDevicesAttrsType[Spi_ExDevicesAttrsType] Spi_JobCfgType --> Spi_JobConfigType_struct[Spi_JobConfigType_struct] </pre>
Data Fields	• Spi_ChannelType tChnlCntOfJob

	Number of channels in the job . • <code>const Spi_ChannelType *pcChnlsArray</code> Channel index list . • <code>Spi_PfnNotifyType pfnJobEndNotify</code> Job end notification. • <code>Spi_PfnNotifyType pfStartNotification</code> Job start notification. • <code>sint8 s8Priority</code> Priority. • <code>uint32 SpiCoreUse</code> CoreID used • <code>Spi_JobStatType *pJobState</code> Implementation-specific field referencing the channel internal state. • <code>Spi_HWUnitType tVirHwModuleIdx</code> HW virtual module. • <code>uint32 u32Hwoffset</code> HW Unit offset. • <code>Spi_ExternalDeviType tExternalDev</code> <code>tExternalDev</code>. • <code>const Spi_ExDevicesAttrsType * pcDeviceAttrs</code> Implementation-specific field: cached LLD device attributes.
--	--

2.5.10 Spi_JobsCfgType

Structure	<code>Spi_JobsCfgType</code>
Description	This is the structure containing Job configuration.



2.5.11 Spi_ChannelbufferType

Structure	Spi_ChannelbufferType
Description	IB/EB Tx & Rx Buffer address.
Diagram	N/A
Data Fields	• Spi_DataBufferType *pSpiTxBuf Tx buffer pointer, const buffer for EB, changeable for IB. • Spi_DataBufferType *pSpiRxBuf Rx buffer pointer.

2.5.12 Spi_ChannelStateType

Structure	Spi_ChannelStateType
Description	Internal structure used to manage the channel state.
Diagram	N/A
Data Fields	• uint8 u8DefaultTransFlag Default Transmit Enabled. • Spi_NumberOfDataType tTransferByteCnt Actual Transfer size for EB.

2.5.13 Spi_ChnlCfgType

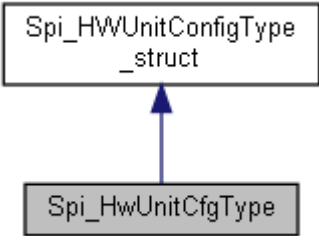
Structure	Spi_ChnlCfgType
Description	The structure contains the channel configuration parameters.
Diagram	<pre> graph TD Spi_ChnlCfgType_struct[Spi_ChnlCfgType_struct] Spi_ChannelbufferType[Spi_ChannelbufferType] Spi_ChannelStateType[Spi_ChannelStateType] Spi_ChnlCfgType[Spi_ChnlCfgType] Spi_ChnlCfgType --> Spi_ChnlCfgType_struct Spi_ChnlCfgType --> Spi_ChannelbufferType Spi_ChnlCfgType --> Spi_ChannelStateType Spi_ChnlCfgType -- pcChnlTxRxBuffer --> Spi_ChannelbufferType Spi_ChnlCfgType -- pChannelState --> Spi_ChannelStateType </pre>
Data Fields	- Spi_IbEbBufferType eBufferType Buffer Type IB/EB. - uint32 u32DefaultTransData Default Transmit Value. - Spi_NumberOfDataType tChnlByteCnt Data length. - Spi_ChannelbufferType *pcChnlTxRxBuffer Buffer Descriptor - uint32 SpiCoreUse CoreID used - Spi_ChannelStateType *pChannelState Implementation-specific field referencing the channel internal state.

2.5.14 Spi_ChannelsCfgType

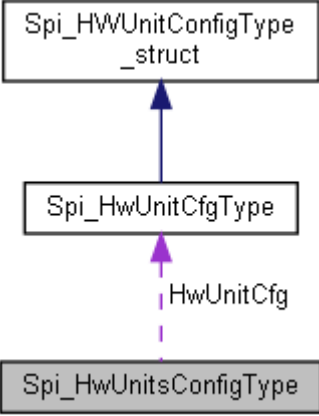
Structure	Spi_ChannelsCfgType
Description	The structure contains the channel configuration
Diagram	<pre> graph TD Spi_ChnlCfgType_struct[Spi_ChnlCfgType_struct] Spi_ChannelbufferType[Spi_ChannelbufferType] Spi_ChannelStateType[Spi_ChannelStateType] Spi_ChnlCfgType[Spi_ChnlCfgType] Spi_ChannelsCfgType[Spi_ChannelsCfgType] Spi_ChannelsCfgType --> Spi_ChnlCfgType_struct Spi_ChannelsCfgType --> Spi_ChnlCfgType Spi_ChannelsCfgType -- ChannelCfg --> Spi_ChnlCfgType Spi_ChnlCfgType --> Spi_ChannelbufferType Spi_ChnlCfgType --> Spi_ChannelStateType Spi_ChnlCfgType -- pcChnlTxRxBuffer --> Spi_ChannelbufferType Spi_ChnlCfgType -- pChannelState --> Spi_ChannelStateType </pre>
Data Fields	const SPI_ChnlCfgType * ChannelCfg

2.5.15 Spi_HwUnitCfgType

Structure	Spi_HwUnitCfgType
Description	This structure holds the hw configuration parameters

Diagram	 <pre> graph BT A[Spi_HwUnitConfigType_struct] --> B[Spi_HwUnitCfgType] </pre>
Data Fields	- uint8 u8HwInstIdx SPI hardware module index of chip - uint32 SpiCoreUse CoreID used - SPI_MasterSlaveModeType eSpiPhyUnitMode Slave Mode - enabled - uint8 u8SyncOrAsync Indicates if the HW unit is configured as Sync or Async - uint16 u16UseDma Boolean flag indicating if DMA will be used or not for this FCSPI unit - uint8 u8TxDmaChannel Master TX DMA channel - enabled by the FCSPI TX source - uint8 u8RxDmaChannel RX DMA channel - enabled by the FCSPI RX source

2.5.16 Spi_HwUnitsConfigType

Structure	Spi_HwUnitsConfigType
Description	This structure holds the PhyUnit configuration
Diagram	 <pre> graph BT A[Spi_HwUnitConfigType_struct] --> B[Spi_HwUnitCfgType] B -- HwUnitCfg --> C[Spi_HwUnitsConfigType] </pre>
Data Fields	const SPI_HwUnitCfgType * HwUnitCfg

2.5.17 Spi_HwScheduleQueueType

Structure	Spi_HwScheduleQueueType
Description	This structure holds the hw scheduling queue.
Diagram	N/A
Data Fields	- sint8 s8MaxSchedJobPriority The highest priority of all the job list - Spi_JobType aSchedJobsFirstNode [SPI_JOB_PRIORITY_LEVELS_COUNT]

	Array of job id in queue to be scheduled, header node for each priority level. - Spi_JobType aSchedJobsLastNode [SPI_JOB_PRIORITY_LEVELS_COUNT] Array of job id in queue to be scheduled, tail node for each priority level. - Spi_StatusType Status FCSPI state.
--	--

2.5.18 Spi_ChnJobSeqHwStateType

Structure	Spi_ChnJobSeqHwStateType
Description	This structure holds the state of each Channel/Job/Sequence/HwUnit
Diagram	<pre> graph TD SCSHT[Spi_ChnJobSeqHwStateType] SSTS[Spi_SeqStateType] SJST[Spi_JobStateType] SCTST[Spi_ChannelStateType] SHSQT[Spi_HwScheduleQueueType] SSCfgT[Spi_SeqCfgType] SSST_struct[Spi_SequenceStateType_struct] SCfg_struct[Spi_SequenceConfigType_struct] SCSHT -.-> aSeqStats SSTS SCSHT -.-> aJobStats SJST SCSHT -.-> aChnlStats SCTST SCSHT -.-> aHwUnitSchedQueue SHSQT SSTS -- pcSeqCfg --> SSCfgT SSTS --> SSST_struct SJST -.-> pAsyncCurSeqStat SSTS SSST_struct --> SCfg_struct </pre>
Data Fields	- Spi_SeqStatType aSeqStats[SPI_SEQUENCE_MAX_COUNT] Sequence state - Spi_JobStatType aJobStats[SPI_JOB_MAX_COUNT] Job state - Spi_ChannelStateType aChnlStats[SPI_CHANNEL_MAX_COUNT] Channel state - Spi_HwScheduleQueueType aHwUnitSchedQueue [SPI_HW_MODULE_MAX_COUNT] Hw scheduling queue

2.5.19 Spi_ConfigType

Structure	Spi_ConfigType
Description	This structure holds the hw configuration parameters

Diagram	
Data Fields	• uint16 u16MaxExternalDevice Number of external devices defined in the configuration • Spi_ChannelType tSpiMaxValidChnl Number of channels defined in the configuration • Spi_JobType tSpiMaxValidJob Number of jobs defined in the configuration • Spi_SequenceType tSpiMaxValidSeq Number of sequences defined in the configuration • uint32 SpiCoreUse CoreID used • const Spi_ChannelsCfgType * ChannelConfig Pointer to Array of channels defined in the configuration • const Spi_JobsCfgType * JobConfig Pointer to Array of jobs defined in the configuration • const SPI_SeqsCfgType * SequenceConfig Pointer to Array of sequences defined in the configuration • const SPI_ChannelsAttrsType * ChannelAttrsConfig Channel attributes • const SPI_ExDevicesAttrsType * ExDeviceConfig FCSPI HW module attributes • const Spi_HwUnitsConfigType * HWUnitConfig Pointer to Array of LLD FCSPi device instances

2.6 API Functions

2.6.1 Functions in Spi.h

2.6.1.1 void Spi_Init(const Spi_ConfigType *ConfigPtr)

Function	Spi_Init(const Spi_ConfigType *ConfigPtr)	
Description	SPI module initialization.	
Diagram	N/A	
Parameters	Parameter	Description
	ConfigPtr	Pointer to configuration set.
Returns	N/A	

2.6.1.2 Std_ReturnType Spi_DeInit(void)

Function	Std_ReturnType Spi_Delnit(void)
-----------------	---------------------------------

Description	This function de-initializes the SPI driver.
Diagram	<pre> graph LR Spi_DelInit --> Spi_GetStatus Spi_DelInit --> Spi_LL_DelInit Spi_GetStatus --> Det_ReportError Spi_LL_DelInit --> FCSPI_HWA_WriteCTRL </pre>
Parameters	N/A
Returns	Std_ReturnType

2.6.1.3 Spi_StatusType Spi_GetStatus(void)

Function	Spi_StatusType Spi_GetStatus(void)
Description	This function returns the status of the SPI driver.
Diagram	N/A
Parameters	N/A
Returns	Spi_StatusType

2.6.1.4 Spi_JobResultType Spi_GetJobResult(Spi_JobType Job)

Function	Spi_JobResultType Spi_GetJobResult(Spi_JobType Job)	
Description	This function is used to request the status of a specific job.	
Diagram	N/A	
Parameters	Parameter	Description
	Job	Job ID
Returns	Spi_JobResultType	

2.6.1.5 Spi_SeqResultType Spi_GetSequenceResult(Spi_SequenceType Sequence)

Function	Spi_SeqResultType Spi_GetSequenceResult(Spi_SequenceType Sequence)	
Description	This service returns the last transmission result of the specified Sequence.	
Diagram	N/A	
Parameters	Parameter	Description
	Sequence	Sequence ID
Returns	Spi_SeqResultType	

2.6.1.6 Std_ReturnType Spi_AsyncTransmit(Spi_SequenceType Sequence)

Function	Std_ReturnType Spi_AsyncTransmit(Spi_SequenceType Sequence)	
Description	This function triggers the asynchronous transmission for the given sequence.	
Diagram	<pre> graph LR Spi_AsyncTransmit --> Spi_LinkJobsToSequence Spi_AsyncTransmit --> Spi_ScheduleJobsOnHwUnit Spi_ScheduleJobsOnHwUnit --> Spi_LL_JobTransfer Spi_LL_JobTransfer --> Spi_LL_Channel_InitDmaRx Spi_LL_JobTransfer --> Spi_LL_Channel_InitDmaTx </pre>	
Parameters	Parameter	Description
	Sequence	Sequence ID
Returns	Std_ReturnType	

2.6.1.7 void Spi_MainFunction_Handling(void)

[illegible]

2.6.1.8 void Spi_AfterOneJobTransferDone (const SPI_JobParamsType *pcJobCfg)

Function	void Spi_AfterOneJobTransferDone(const SPI_JobParamsType *pcJobCfg)
Description	This function is called after a Job has been executed.

Diagram	<pre>graph LR SPl_AlterOneJobTransferDone[SPl_AlterOneJobTransferDone] --> SchM_Enter_SPI_03[SchM_Enter_Spi_SPI_EXCLUSIVE_AREA_03] SPl_AlterOneJobTransferDone --> SchM_Enter_SPI_04[SchM_Enter_Spi_SPI_EXCLUSIVE_AREA_04] SPl_AlterOneJobTransferDone --> SchM_Enter_SPI_06[SchM_Enter_Spi_SPI_EXCLUSIVE_AREA_06] SPl_AlterOneJobTransferDone --> SchM_Enter_SPI_07[SchM_Enter_Spi_SPI_EXCLUSIVE_AREA_07] SPl_AlterOneJobTransferDone --> SchM_Exit_SPI_03[SchM_Exit_Spi_SPI_EXCLUSIVE_AREA_03] SPl_AlterOneJobTransferDone --> SchM_Exit_SPI_04[SchM_Exit_Spi_SPI_EXCLUSIVE_AREA_04] SPl_AlterOneJobTransferDone --> SchM_Exit_SPI_06[SchM_Exit_Spi_SPI_EXCLUSIVE_AREA_06] SPl_AlterOneJobTransferDone --> SchM_Exit_SPI_07[SchM_Exit_Spi_SPI_EXCLUSIVE_AREA_07] SPl_AlterOneJobTransferDone --> SPl_ScheduleJobsOnHwUnit[SPl_ScheduleJobsOnHwUnit] SPl_AlterOneJobTransferDone --> SPl_StartFirstJobScheduleNext[SPl_StartFirstJobScheduleNext] SPl_AlterOneJobTransferDone --> SPl_UnlinkRemainJobsInSeq[SPl_UnlinkRemainJobsInSeq] SPl_ScheduleJobsOnHwUnit --> SPl_LL_JobTransfer[SPl_LL_JobTransfer] SPl_StartFirstJobScheduleNext --> SPl_LL_JobTransfer SPl_UnlinkRemainJobsInSeq --> SPl_LL_JobTransfer SPl_LL_JobTransfer --> FCSPi_HwA_GetFifoStatus[FCSPi_HwA_GetFifoStatus] SPl_LL_JobTransfer --> FCSPi_HwA_SetCTRL[FCSPi_HwA_SetCTRL] SPl_LL_JobTransfer --> FCSPi_HwA_WriteCLK_CFG[FCSPi_HwA_WriteCLK_CFG] SPl_LL_JobTransfer --> SPl_LL_Channel_InitDmaRx[SPl_LL_Channel_InitDmaRx] SPl_LL_JobTransfer --> SPl_LL_Channel_InitDmaTx[SPl_LL_Channel_InitDmaTx] SPl_LL_Channel_InitDmaTx --> Dma_SetCfGdAddr[Dma_SetCfGdAddr] SPl_LL_Channel_InitDmaTx --> Dma_SetCfGdOff[Dma_SetCfGdOff] SPl_LL_Channel_InitDmaTx --> Dma_SetCfGSaddr[Dma_SetCfGSaddr] SPl_LL_Channel_InitDmaTx --> Dma_SetCfGSoff[Dma_SetCfGSoff] SPl_LL_Channel_InitDmaTx --> Dma_UpdateOuterLoopCounter[Dma_UpdateOuterLoopCounter] SPl_LL_Channel_InitDmaTx --> Dma_EnableHwRequest[Dma_EnableHwRequest] SPl_LL_Channel_InitDmaTx --> FCSPi_HwA_ClearDMAEnableRegbit[FCSPi_HwA_ClearDMAEnableRegbit] SPl_LL_Channel_InitDmaTx --> FCSPi_HwA_SetFOWTm[FCSPi_HwA_SetFOWTm] SPl_LL_Channel_InitDmaTx --> FCSPi_HwA_SetTrRxControl[FCSPi_HwA_SetTrRxControl] SPl_LL_Channel_InitDmaTx --> Dma_LL_SetDADDR[Dma_LL_SetDADDR] SPl_LL_Channel_InitDmaTx --> Dma_LL_SetChannelLoopOffsetAndBYTES[Dma_LL_SetChannelLoopOffsetAndBYTES] SPl_LL_Channel_InitDmaTx --> Dma_LL_SetModuleAndSize_1[Dma_LL_SetModuleAndSize] SPl_LL_Channel_InitDmaTx --> Dma_LL_SetDOFF[Dma_LL_SetDOFF] SPl_LL_Channel_InitDmaTx --> Dma_LL_SetSADDR[Dma_LL_SetSADDR] SPl_LL_Channel_InitDmaTx --> Dma_LL_SetModuleAndSize_2[Dma_LL_SetModuleAndSize] SPl_LL_Channel_InitDmaTx --> Dma_LL_SetSOFF[Dma_LL_SetSOFF] SPl_LL_Channel_InitDmaTx --> Dma_LL_UpdateOuterLoopCounter[Dma_LL_UpdateOuterLoopCounter] SPl_LL_Channel_InitDmaTx --> Dma_LL_EnableHwRequest[Dma_LL_EnableHwRequest] Dma_LL_SetChannelLoopOffsetAndBYTES --> Dma_ValidateCommonPointer[Dma_ValidateCommonPointer] Dma_LL_SetModuleAndSize_1 --> Dma_ValidateCommonPointer Dma_LL_SetModuleAndSize_2 --> Dma_ValidateChannelIndex[Dma_ValidateChannelIndex] Dma_LL_SetSOFF --> Dma_ValidateChannelIndex Dma_LL_UpdateOuterLoopCounter --> Dma_ValidateChannelIndex Dma_LL_EnableHwRequest --> Dma_ValidateChannelIndex Dma_ValidateCommonPointer --> Det_ReportError[Det_ReportError] Dma_ValidateChannelIndex --> Det_ReportError</pre>	
Parameters	Parameter	Description
	pcJobCfg	
Returns	N/A	

2.6.1.9 void Spi_GetVersionInfo (Std_VersionInfoType *versioninfo)

Function	void Spi_GetVersionInfo (Std_VersionInfoType *versioninfo)	
Description	Returns the version information of this SPI driver module.	
Diagram	N/A	
Parameters	Parameter	Description
	versioninfo	SPI module version information value pointer.
Returns	N/A	

2.6.1.10 Std_ReturnType Spi_WriteB(Spi_ChannelType Channel, const Spi_DataBufferType *DataBufferPtr)

Function	Std_ReturnType Spi_WriteIB(Spi_ChannelType Channel, const Spi_DataBufferType *DataBufferPtr)	
Description	Writes the given data into the buffer of a specific channel.	
Diagram	N/A	
Parameters	Parameter	Description
	Channel	Channel ID.
	DataBufferPtr	Pointer to source data buffer, if NULL, use default data.
Returns	Std_ReturnType	

2.6.1.11 Std_ReturnType Spi_ReadIB(Spi_ChannelType Channel, Spi_DataBufferType *DataBufferPointer)

Function	Std_ReturnType Spi_ReadIB(Spi_ChannelType Channel, Spi_DataBufferType *DataBufferPointer)
Description	This function reads the data from the buffer of a channel and puts in the memory location.
Diagram	N/A

Parameters	Parameter	Description
	Channel	Channel ID.
	DataBufferPointer	Pointer to the memory location that will be written with the data in the internal buffer.
Returns	Std_ReturnType	

2.6.1.12 Std_ReturnType Spi_SetupEB(Spi_ChannelType Channel, const Spi_DataBufferType *SrcDataBufferPtr, Spi_DataBufferType *DesDataBufferPtr, Spi_NumberOfDataType Length)

Function	Std_ReturnType Spi_SetupEB(Spi_ChannelType Channel, const Spi_DataBufferType *SrcDataBufferPtr, Spi_DataBufferType *DesDataBufferPtr, Spi_NumberOfDataType Length)	
Description	This function setup an external buffer to be used by a specific channel.	
Diagram	N/A	
Parameters	Parameter	Description
	Channel	Channel ID.
	SrcDataBufferPtr	Pointer to the memory location that will hold the transmitted data.
	Length	Length of the data in the external buffer.
	DesDataBufferPtr	Pointer to the memory location that will hold the received data.
Returns	Std_ReturnType	

2.6.1.13 Std_ReturnType Spi_SyncTransmit(Spi_SequenceType Sequence)

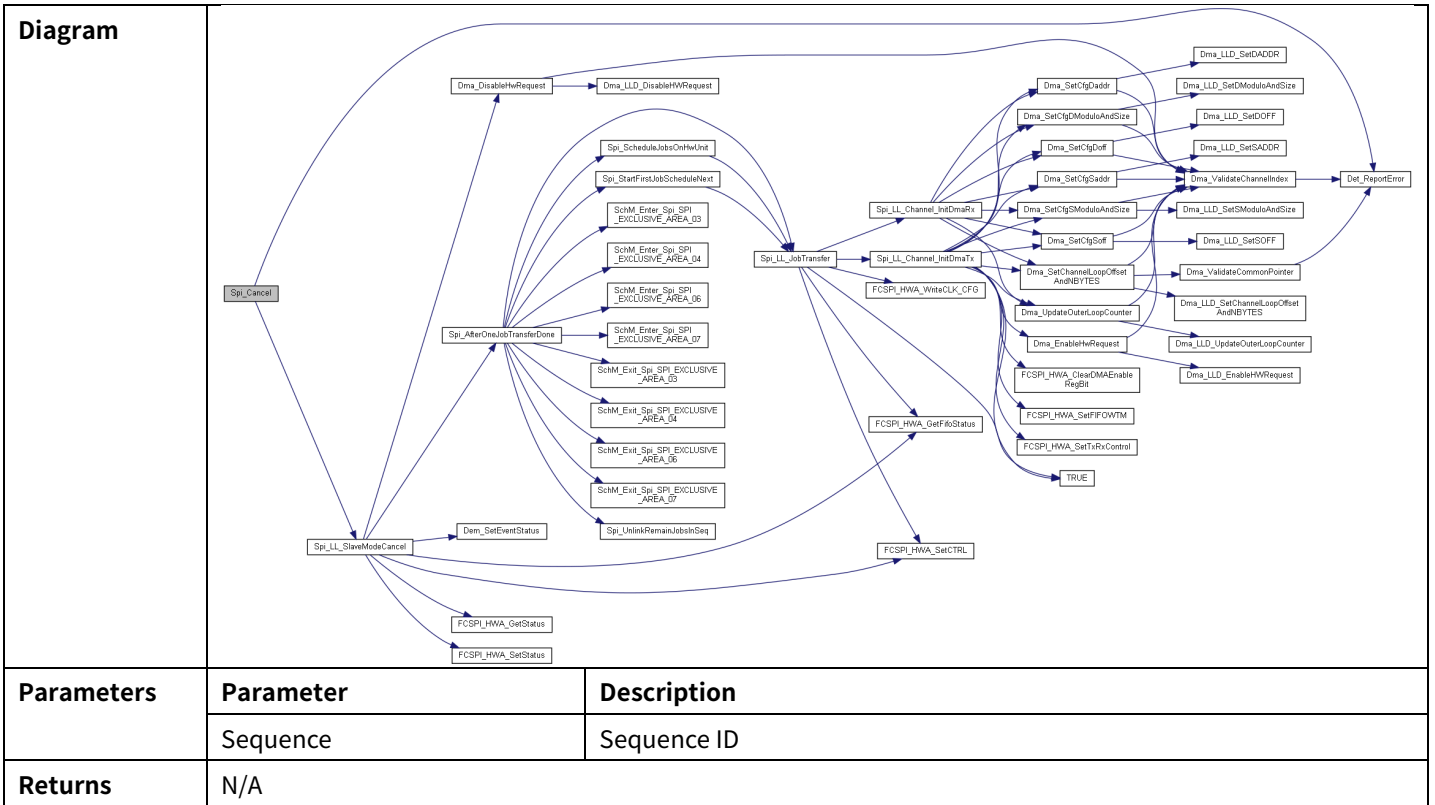
Function	Std_ReturnType Spi_SyncTransmit(Spi_SequenceType Sequence)	
Description	Service to transmit data of sequence on the SPI bus synchronously.	
Diagram	N/A	
Parameters	Parameter	Description
	Sequence	Sequence ID
Returns	Std_ReturnType	

2.6.1.14 Spi_StatusType Spi_GetHWUnitStatus(Spi_HWUnitType HWUnit)

Function	Spi_StatusType Spi_GetHWUnitStatus(Spi_HWUnitType HWUnit)	
Description	This function is used to request the status of a specific SPI peripheral unit.	
Diagram	N/A	
Parameters	Parameter	Description
	HWUnit	The HW peripheral for which we need the status
Returns	Spi_StatusType	

2.6.1.15 void Spi_Cancel(Spi_SequenceType Sequence)

Function	void Spi_Cancel(Spi_SequenceType Sequence)	
Description	This function is used to request the cancelation of the given sequence.	



2.6.1.16 Std_ReturnType Spi_SetAsyncMode(Spi_AsyncModeType AsyncMode)

Function	Std_ReturnType Spi_SetAsyncMode(Spi_AsyncModeType AsyncMode)	
Description	This function specifies the asynchronous mode for the SPI busses handled asynchronously.	
Diagram	N/A	
Parameters	Parameter	Description
	AsyncMode	The asynchronous operating mode (SPI_POLLING_MODE or SPI_INTERRUPT_MODE)
Returns	Std_ReturnType	

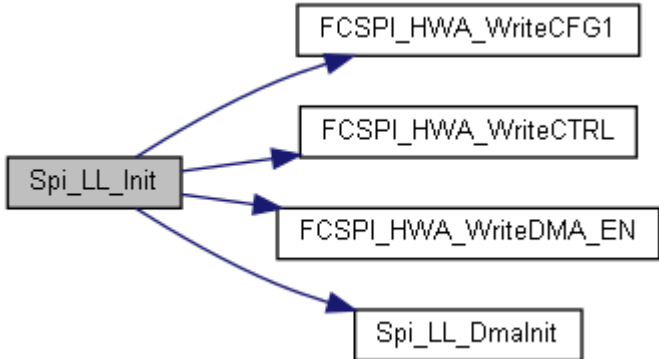
2.6.1.17 Std_ReturnType SPI_SetHwAsyncMode(Spi_HWUnitType tHwUnit, Spi_AsyncModeType AsyncMode)

Function	Std_ReturnType SPI_SetHwAsyncMode(Spi_HWUnitType tHwUnit, Spi_AsyncModeType AsyncMode)	
Description	This function specifies the asynchronous mode for a given HWUnit	
Diagram	N/A	
Parameters	Parameter	Description
	tHwUnit	The ID of the hardware to be configured
	AsyncMode	This parameter specifies the asynchronous operating mode (SPI_POLLING_MODE or SPI_INTERRUPT_MODE)
Returns	Std_ReturnType	

2.7 Driver Functions

2.7.1 Functions in Spi_FCSpi.h

2.7.1.1 void Spi_LL_Init(Spi_HWUnitType tVirHwUnit, uint8 SpiCoreID)

Function	void Spi_LL_Init(Spi_HWUnitType tVirHwUnit, uint8 SpiCoreID)	
Description	Initialize FCSPI unit in low level.	
Diagram	 <pre> graph LR A[Spi_LL_Init] --> B[FCSPI_HWA_WriteCFG1] A --> C[FCSPI_HWA_WriteCTRL] A --> D[FCSPI_HWA_WriteDMA_EN] A --> E[Spi_LL_DmaInit] </pre>	
Parameters	Parameter	Description
	tVirHwUnit	Specifies which FCSPI peripheral is used
	SpiCoreID	Target core to be operated
Returns	N/A	
Referenced By	Spi_Init	

2.7.1.2 void Spi_LL_DeInit(Spi_HWUnitType tVirHwUnit, uint8 SpiCoreID)

Function	void Spi_LL_DeInit(Spi_HWUnitType tVirHwUnit)	
Description	De-initialize FCSPI Hw unit.	
Diagram	N/A	
Parameters	Parameter	Description
	tVirHwUnit	Specifies which FCSPI peripheral is used
	SpiCoreID	The ID of the hardware to be configured
Returns	N/A	

2.7.1.3 void Spi_LL_JobTransfer (const Spi_JobCfgType * pcJobCfg)

Function	void Spi_LL_JobTransfer (const Spi_JobCfgType * pcJobCfg)
Description	This function sets the Job's attributes and send all the data through the SPI bus.

Diagram		
Parameters	Parameter	Description
	pcJobCfg	Specifies the Job for which we will set the register values
Returns	N/A	

2.7.1.4 void Spi_LL_IrqConfig(Spi_HWUnitType tVirHwUnit, Spi_AsyncModeType AsyncMode)

Function	void Spi_LL_IrqConfig(Spi_HWUnitType tVirHwUnit, Spi_AsyncModeType AsyncMode)	
Description	Activate/deactivate TX_FIFO_Interrupt , RX_FIFO_Interrupt for async hw according to he driver operation mode (polling or interrupt).	
Diagram		
Parameters	Parameter	Description
	tVirHwUnit	Specifies which FCSPi peripheral is used
	AsyncMode	Specifies the async mode of this Spi Hw unit (polling or interrupt).
Returns	N/A	

2.7.1.5 void Spi_LL_IrqPoll(Spi_HWUnitType tVirHwUnit)

Function	void Spi_LL_IrqPoll(Spi_HWUnitType tVirHwUnit)
Description	IRQ polling functions.


```

graph TD
    Spi_LL_IsrPoll --> Dma_CheckTransferCompleted
    Spi_LL_IsrPoll --> Dma_ClearDoneFlag
    Spi_LL_IsrPoll --> Spi_LL_IsrDmaTx
    Spi_LL_IsrPoll --> Spi_LL_IsrDmaRx
    Spi_LL_IsrPoll --> Spi_ScheduleJobsOehwUnit
    Spi_LL_IsrPoll --> SCHM_Exit_Spi_SPI_EXCLUSIVE_AREA_04
    Spi_LL_IsrPoll --> SCHM_Exit_Spi_SPI_EXCLUSIVE_AREA_05
    Spi_LL_IsrPoll --> Spi_StartFirstJobScheduleNext
    Spi_LL_IsrPoll --> Spi_LL_IsrRxBx
    Spi_LL_IsrDmaTx --> Spi_LL_Job_RestartDmaTxTransfer
    Spi_LL_IsrDmaRx --> Spi_LL_Job_RestartDmaRxTransfer
    Spi_LL_IsrDmaTx --> Spi_LL_Channel_InitDmaTx
    Spi_LL_IsrDmaRx --> Spi_LL_Channel_InitDmaRx
    Spi_LL_IsrDmaTx --> FCSPI_HWA_SetCtrl
    Spi_LL_IsrDmaRx --> FCSPI_HWA_SetCtrl
    Spi_LL_IsrDmaTx --> FCSPI_HWA_GetFileStatus
    Spi_LL_IsrDmaRx --> FCSPI_HWA_GetFileStatus
    Spi_LL_IsrDmaTx --> Dma_StartChannel
    Spi_LL_IsrDmaRx --> Dma_StartChannel
    Spi_LL_IsrDmaTx --> Dma_EnableHwRequest
    Spi_LL_IsrDmaRx --> Dma_EnableHwRequest
    Spi_LL_IsrDmaTx --> Dma_UpdateOuterLoopCounter
    Spi_LL_IsrDmaRx --> Dma_UpdateOuterLoopCounter
    Spi_LL_IsrDmaTx --> Dma_SetCtgSof
    Spi_LL_IsrDmaRx --> Dma_SetCtgSof
    Spi_LL_IsrDmaTx --> Dma_SetCtgDoff
    Spi_LL_IsrDmaRx --> Dma_SetCtgDoff
    Spi_LL_IsrDmaTx --> Dma_SetCtgDaddr
    Spi_LL_IsrDmaRx --> Dma_SetCtgDaddr
    Spi_LL_IsrDmaTx --> Dma_SetCtgSaddr
    Spi_LL_IsrDmaRx --> Dma_SetCtgSaddr
    Spi_LL_IsrDmaTx --> Dma_VldateChannelIndex
    Spi_LL_IsrDmaRx --> Dma_VldateChannelIndex
    Spi_LL_IsrDmaTx --> Det_ReportError
    Spi_LL_IsrDmaRx --> Det_ReportError

```

The diagram illustrates the execution flow of the `Spi_LL_IsrPoll` function. It begins by checking for transfer completion and clearing done flags, then proceeds to handle DMA transactions. The core logic involves initializing DMA channels, scheduling jobs, and managing hardware state through various peripheral control registers and state machines. The process concludes with channel validation and error reporting.

2.7.1.6 void Spi_LL_IsrFifoRx (Spi_DeviceStateType * pDevState, uint8 u8HwInstIdx)

Function	void Spi_LL_IsrFifoRx (Spi_DeviceStateType * pDevState, uint8 u8HwInstIdx)	
Description	This function is called by TDF or RDF ISRs for the FCSPI units working in FIFO mode	
Diagram		
Parameters	Parameter	Description
	pDevState	Specifies which FCSPI peripheral is used
	u8HwInstIdx	Specifies which Hw instance is used
Returns	N/A	

2.7.1.7 void Spi_LL_IsrDmaTx (Spi_DeviceStateType * pDevState)

Function	void Spi_LL_IsrDmaTx (Spi_DeviceStateType * pDevState)	
Description	This function is called by DMA ISR for Rx DMA channels.	
Diagram		
Parameters	Parameter	Description
	pDevState	Specifies which FCSPI peripheral is used
Returns	N/A	

2.7.1.8 void Spi_LL_IsrDmaRx (Spi_DeviceStateType * pDevState)

Function	void Spi_LL_IsrDmaRx (Spi_DeviceStateType * pDevState)	
Description	This function is called by DMA ISR for Rx DMA channels.	
Diagram		
Parameters	Parameter	Description
	pDevState	Specifies which FCSPI peripheral is used
Returns	N/A	

	pDevState	Specifies which FCSPI peripheral is used
Returns	N/A	

2.7.1.9 void Spi_LL_SlaveModeCancel (Spi_SequenceType Sequence)

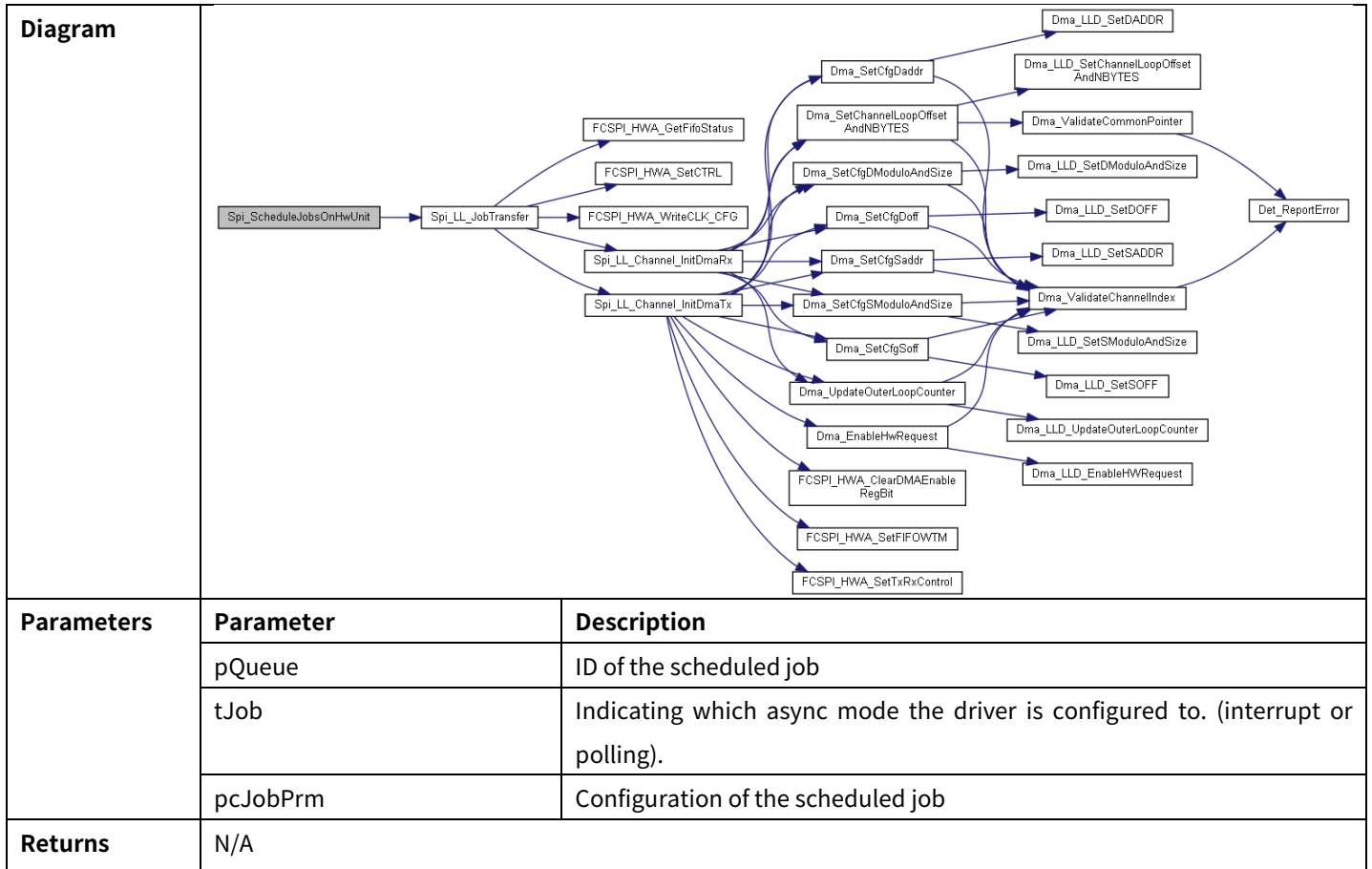
Function	void Spi_LL_SlaveModeCancel (Spi_SequenceType Sequence)	
Description	Cancel the sequence of fcspi that working asynchronously in slave mode.	
Diagram		
Parameters	Parameter	Description
	Sequence	Specifies which sequence is going to be canceled.
Returns	N/A	

2.7.1.10 Std_ReturnType Spi_LL_SyncTransfer (Spi_SequenceType Sequence , uint8 SpiCoreID)

Function	Std_ReturnType Spi_LL_SyncTransfer (Spi_SequenceType Sequence , uint8 SpiCoreID)	
Description	Transfer a sequence synchronously.	
Diagram		
Parameters	Parameter	Description
	pDevState	Specifies which FCSPI peripheral is used
	SpiCoreID	The ID of the hardware to be configured
Returns	Std_ReturnType	

2.7.1.11 void Spi_ScheduleJobsOnHwUnit(Spi_HwScheduleQueueType *pQueue, Spi_JobType tJob, const Spi_JobCfgType *pcJobPrm)

Function	void Spi_ScheduleJobsOnHwUnit(Spi_HwScheduleQueueType *pQueue, Spi_JobType tJob, const Spi_JobCfgType *pcJobPrm)	
Description	This function will schedule a job for a given HW unit.	

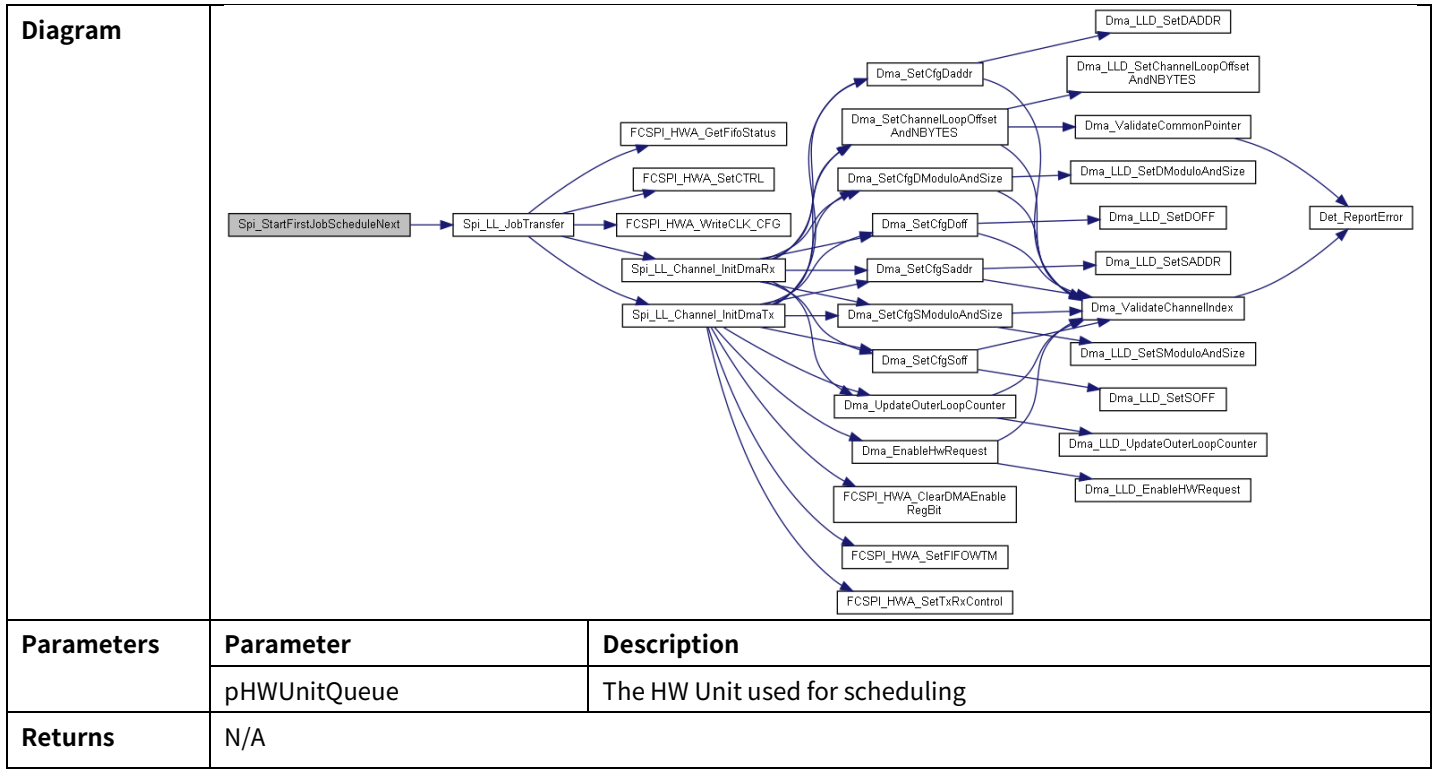


2.7.1.12 Std_ReturnType Spi_LinkJobsToSequence(Spi_SequenceType tSequence, const Spi_SeqCfgType *pcSeq)

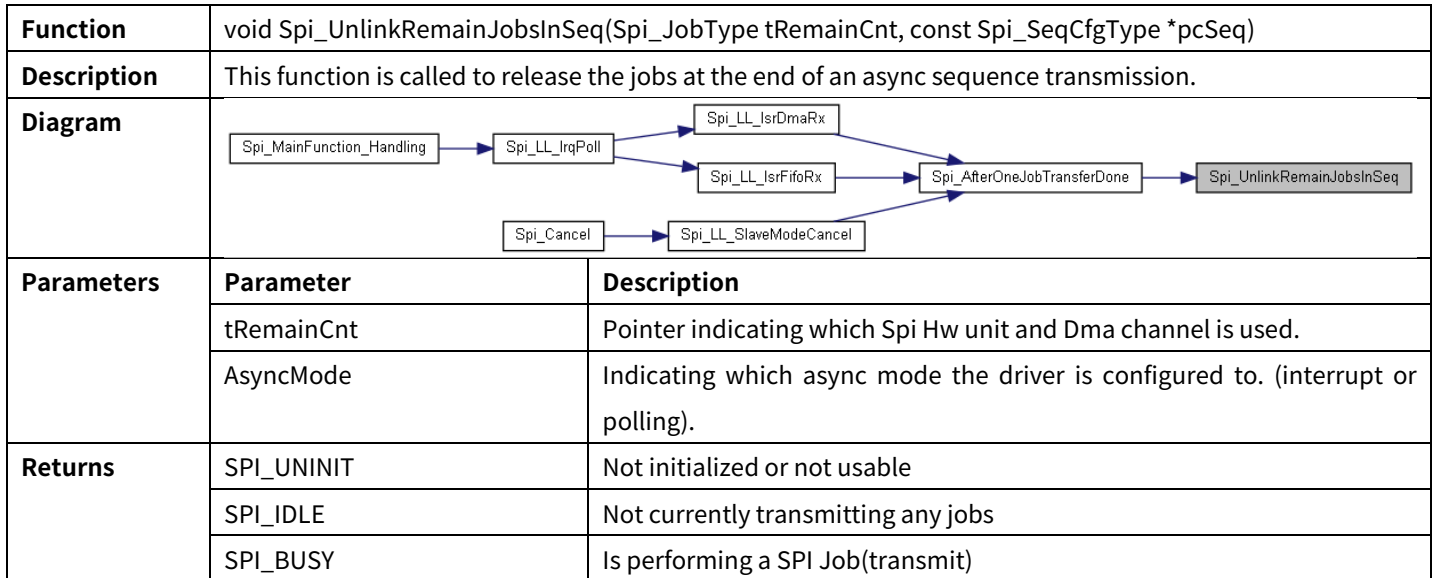
Function	Std_ReturnType Spi_LinkJobsToSequence(Spi_SequenceType tSequence, const Spi_SeqCfgType *pcSeq)	
Description	This function is called in order to mark the jobs of a sequence as ready to be transmitted.	
Diagram	N/A	
Parameters	Parameter	Description
	tSequence	The sequence ID.
	pcSeq	The sequence configuration
Returns	E_OK	The given sequence does not share its jobs with some other sequences, and all its jobs were successfully linked.
	E_NOT_OK	The given sequence shares its jobs with some other sequences. No lock performed for its jobs.

2.7.1.13 void Spi_StartFirstJobScheduleNext(Spi_HwScheduleQueueType *pHWUnitQueue)

Function	void Spi_StartFirstJobScheduleNext(Spi_HwScheduleQueueType *pHWUnitQueue)
Description	This function starts the transfer of the first scheduled job for a given HW unit.

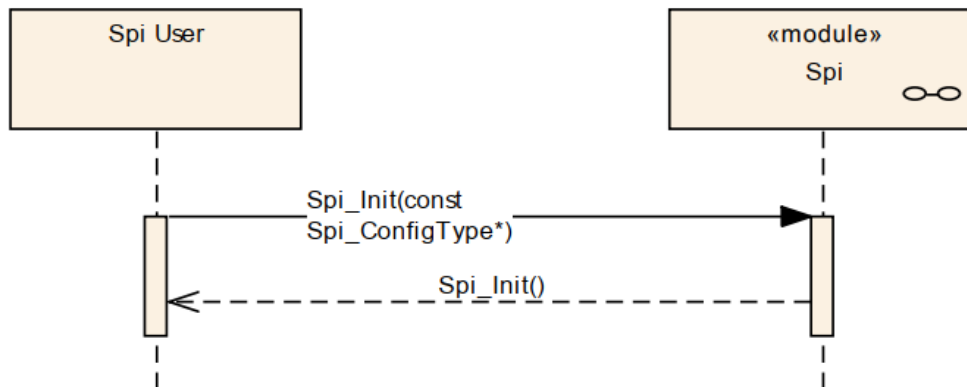


2.7.1.14 void Spi_UnlinkRemainJobsInSeq(Spi_JobType tRemainCnt, const Spi_SeqCfgType *pcSeq)



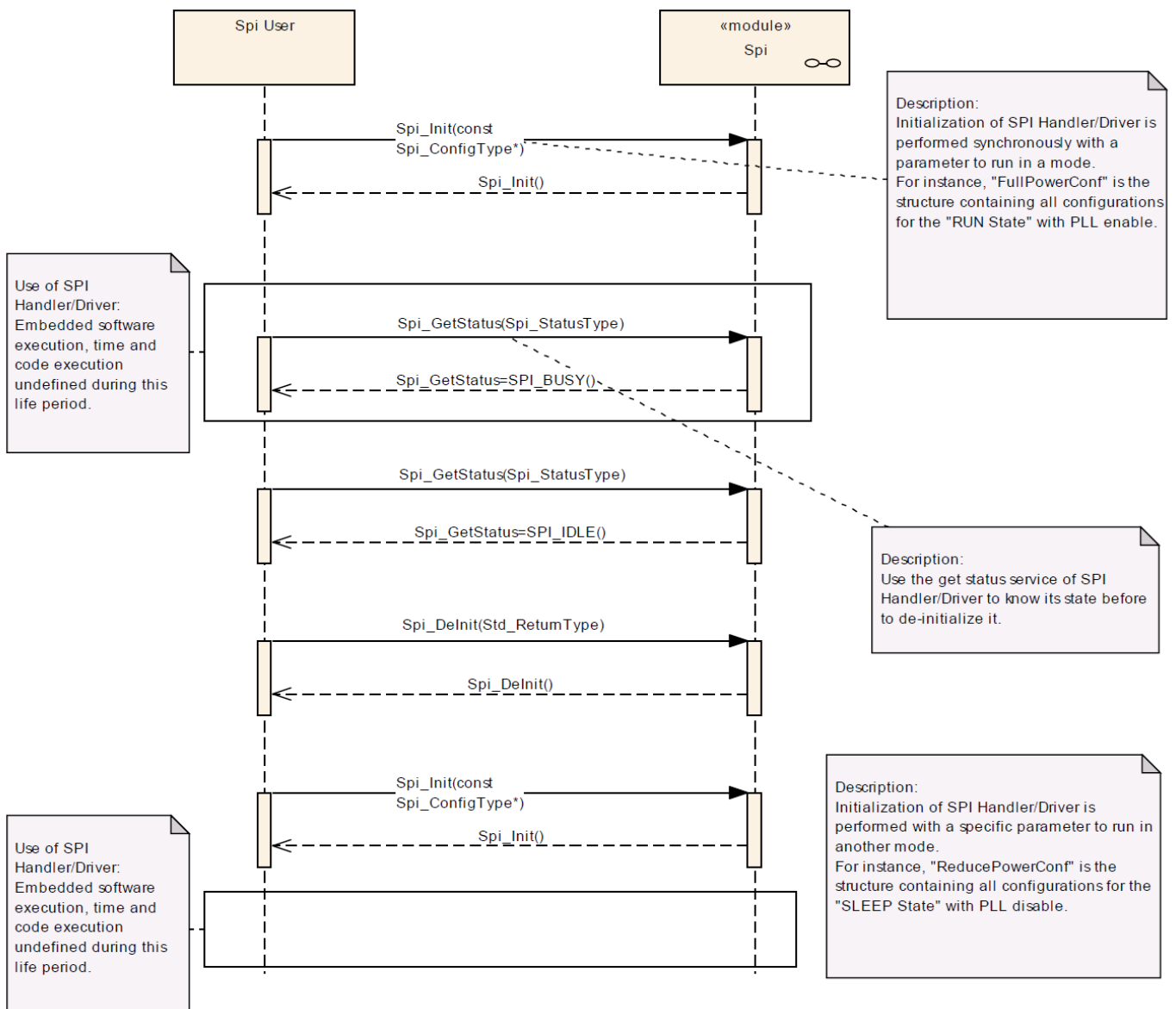
2.8 API Sequence Diagram

2.8.1 Initialization



2.8.2 Modes Transitions

The following sequence diagram shows an example of an Init / DeInit call for a running mode transition.

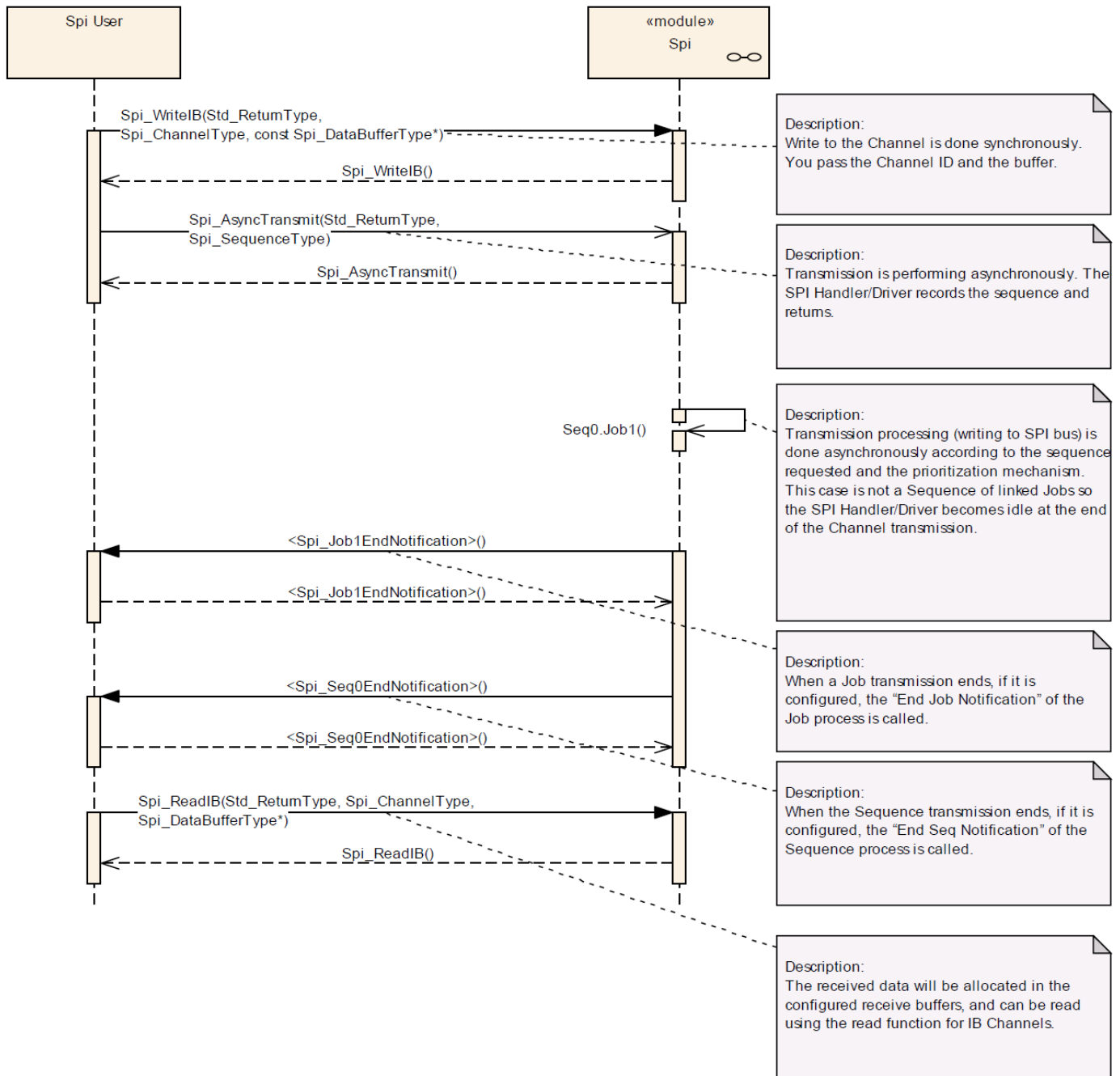


2.8.3 Write/AsyncTransmit/Read (IB)

2.8.3.1 One Channel, One Job Then One Sequence

The following sequence diagram shows an example of Spi_WriteIB / Spi_AsyncTransmit / Spi_ReadIB calls for a Sequence transmission with only one Job composed of only one Channel. Write or Read steps could be skipped when Job is just reading or writing respectively.

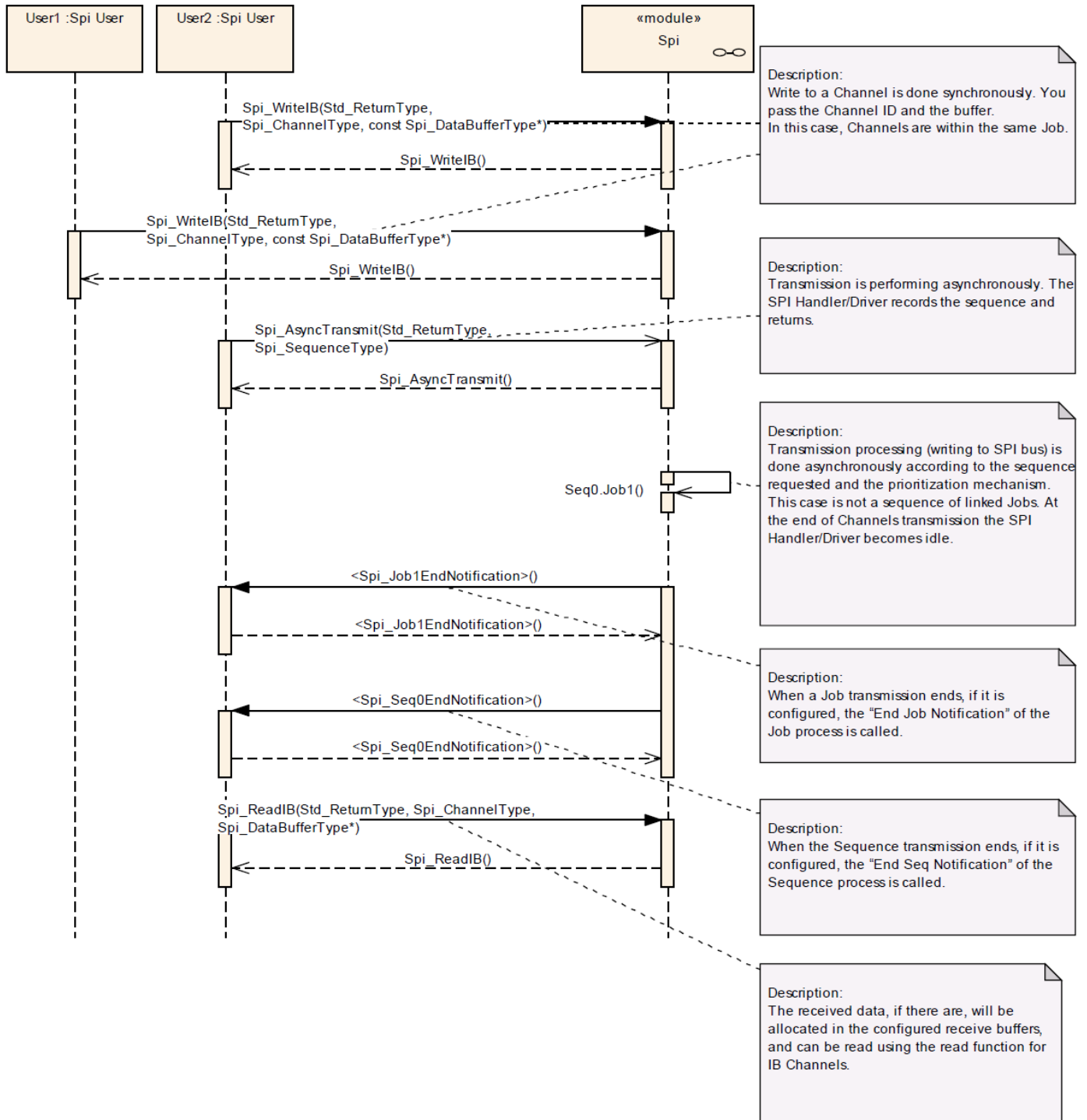
Example: Channel ID 2 belongs to Job ID 1 which belongs to Sequence ID 0.



2.8.3.2 Many Channels, One Job Then One Sequence

The following sequence diagram shows an example of Spi_WriteIB / Spi_AsyncTransmit / Spi_ReadIB calls for a Sequence transmission with only one Job composed of many Channels. Write or Read steps could be skipped when Job is just reading or writing respectively.

Example: Channels ID 2 & 3 belong to Job ID 1 which belongs to Sequence ID 0.



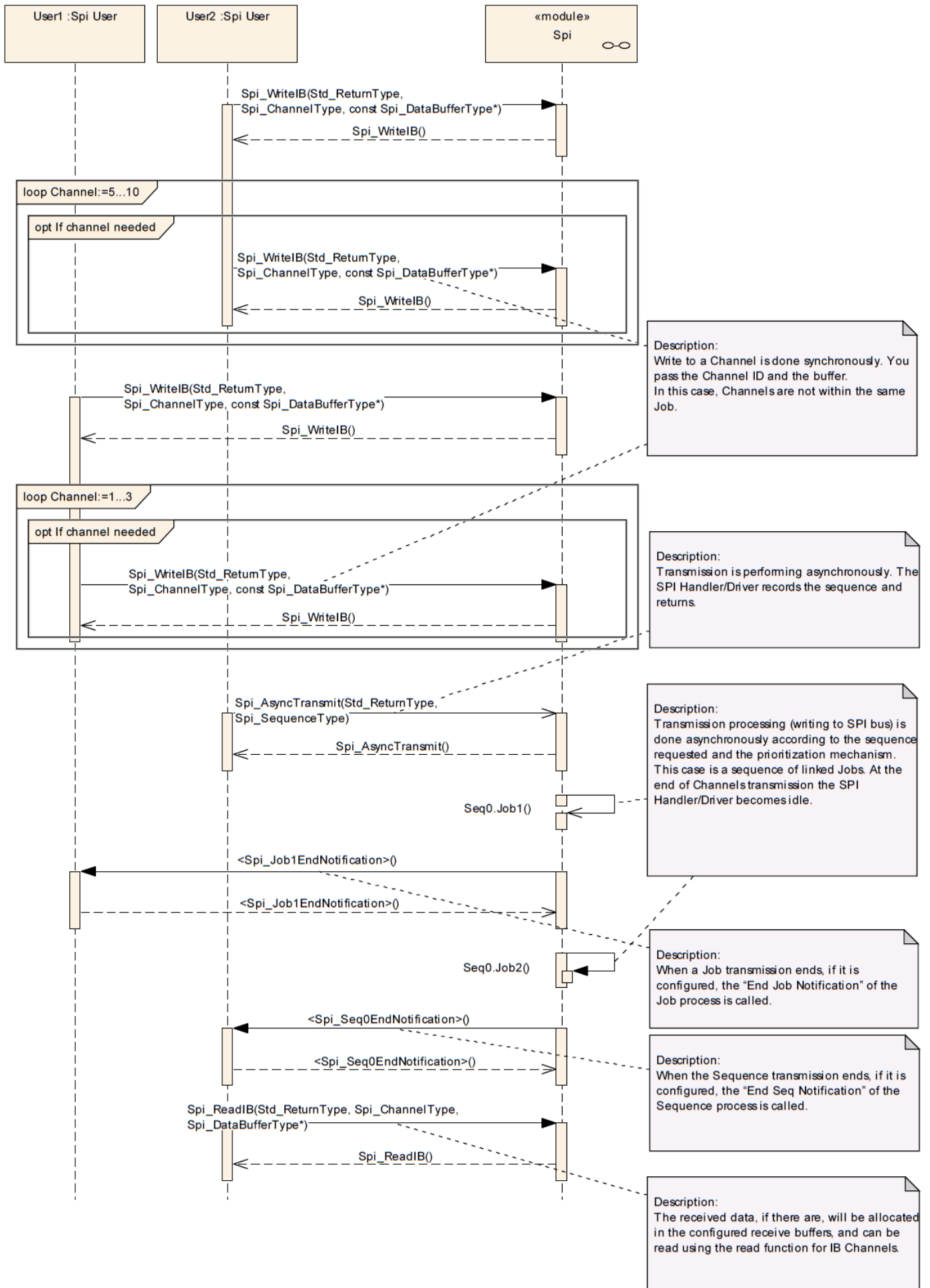
2.8.3.3 Many Channels, Many Jobs and One Sequence

The following sequence diagram shows an example of Spi_WriteIB / Spi_AsyncTransmit / Spi_ReadIB calls for a Sequence transmission of linked Jobs.

Write or Read steps could be skipped when Jobs are just reading or writing respectively.

Example: Channels ID 0 to 3 belong to Job ID 1 (higher priority), Channels ID 4 to 10 belong to Job ID 2 (Lower priority) which has not an end notification function. These Jobs belong to the same Sequence ID 0.

Sequence	Job		Channel
	Name	Priority	
ID0	ID1	High	ID0...ID3
	ID2	Low	ID4...ID10



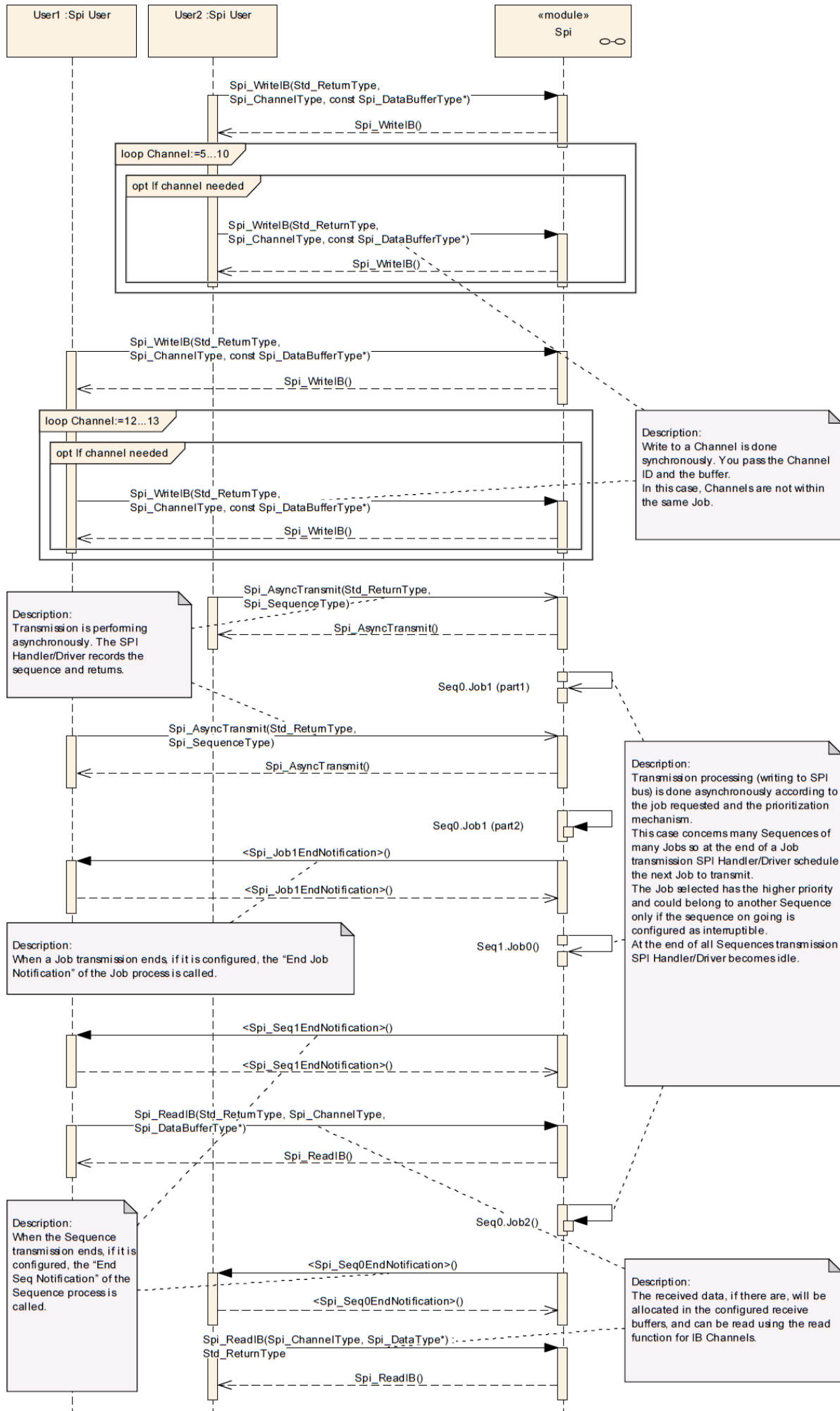
2.8.3.4 Many Channels, Many Jobs and Many Sequences

The following sequence diagram shows an example of Spi_WriteIB / Spi_AsyncTransmit / Spi_ReadIB calls for Sequences transmission. Write or Read steps could be skipped when Jobs are just reading or writing respectively.

Example: Channels ID 0 to 3 belong to Job ID 1 (high priority 2), Channels ID 4 to 10 belong to Job ID 2 (Low priority 1) which has not an end notification function. These Jobs belong to the same Sequence ID 0 which is configured as interruptible.

Channels ID 11 to 13 belong to Job ID 0 (higher priority 3) which belongs to Sequence ID 1 which is configured as not interruptible.

Sequence		Job		Channel
Name	Interruptible	Name	Priority	
ID0	Yes	ID1	2	ID0...ID3
		ID2	1	ID4...ID10
ID1	No	ID0	3	ID11...ID13

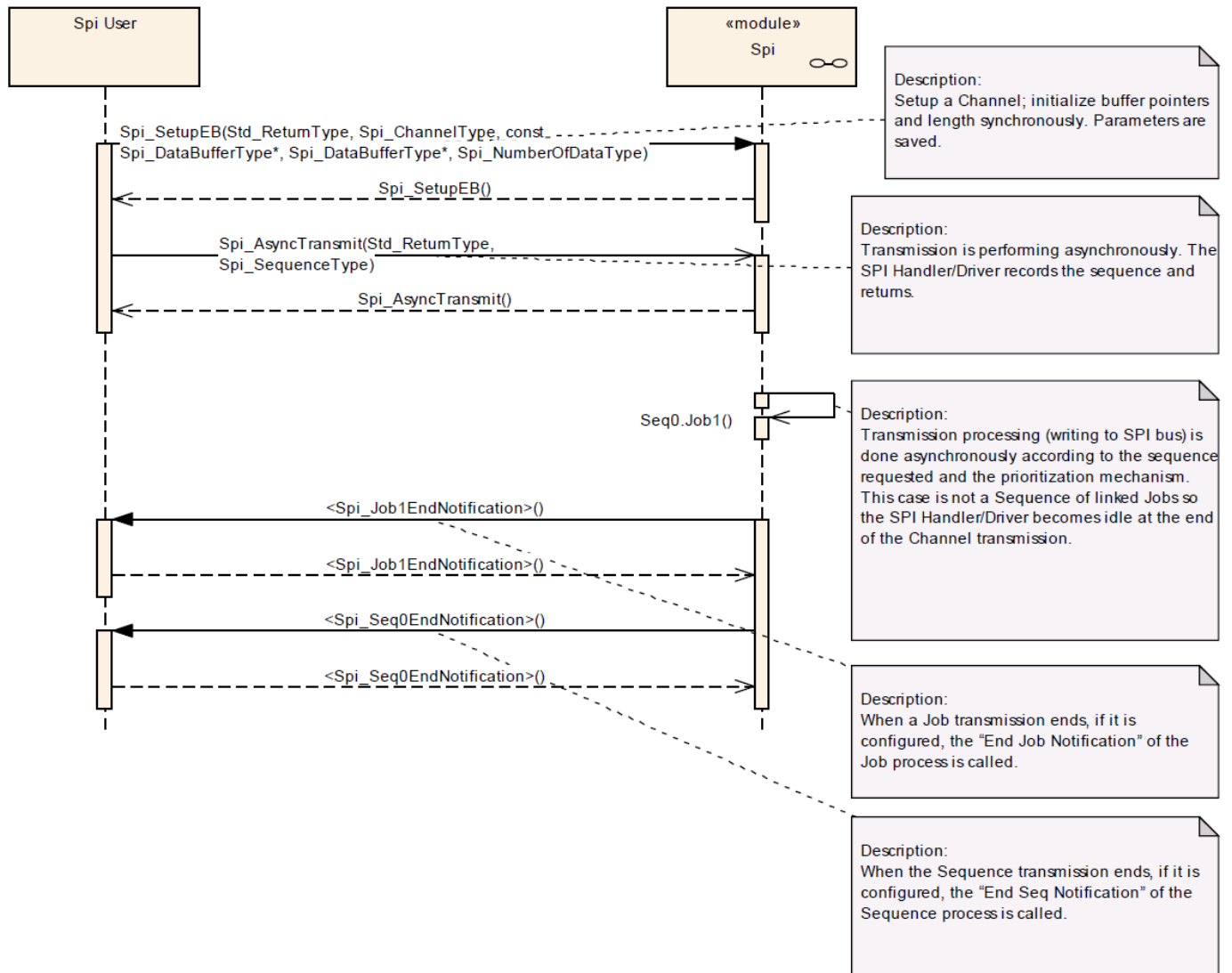


2.8.4 Setup/AsyncTransmit (EB)

2.8.4.1 One Channel, One Job Then One Sequence

The following sequence diagram shows an example of Spi_SetupEB / Spi_AsyncTransmit calls for a Sequence transmission with only one Job composed of only one Channel. Write or Read accesses are "User Dependent" and could be skipped when Job is just reading or writing respectively.

Example: Channel ID 2 belongs to Job ID 1 which belongs to Sequence ID 0.

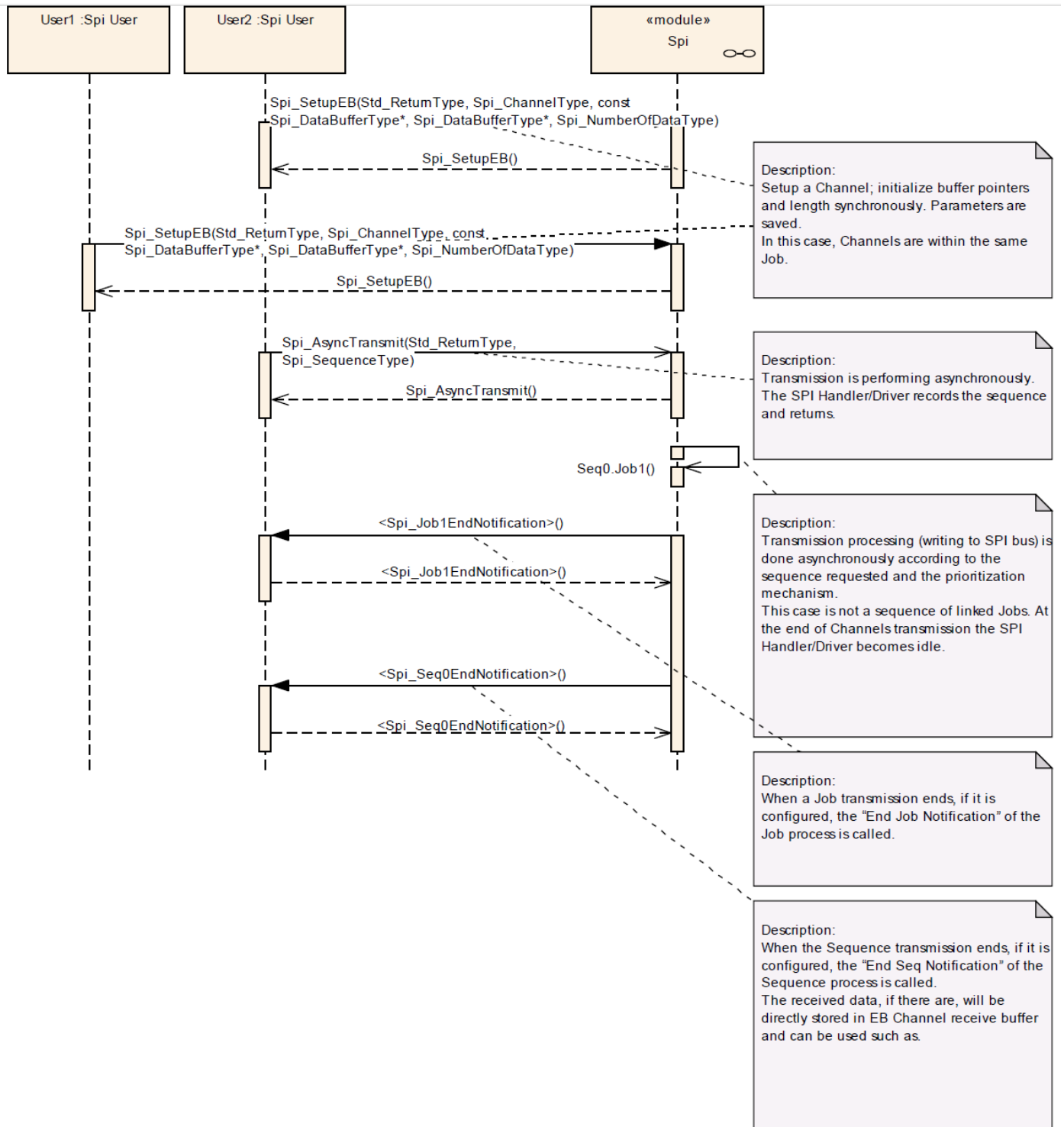


2.8.4.2 Many Channels, One Job Then One Sequence

The following sequence diagram shows an example of Spi_SetupEB / Spi_AsyncTransmit calls for a Sequence transmission with only one Job composed of many Channels. Write or Read accesses are "User Dependent" and could be skipped when Job is just reading or writing respectively.

Example: Channels ID 2 & 3 belong to Job ID 1 which belongs to Sequence ID 0.

Sequence	Job	Channel
ID0	ID1	ID2
		ID3

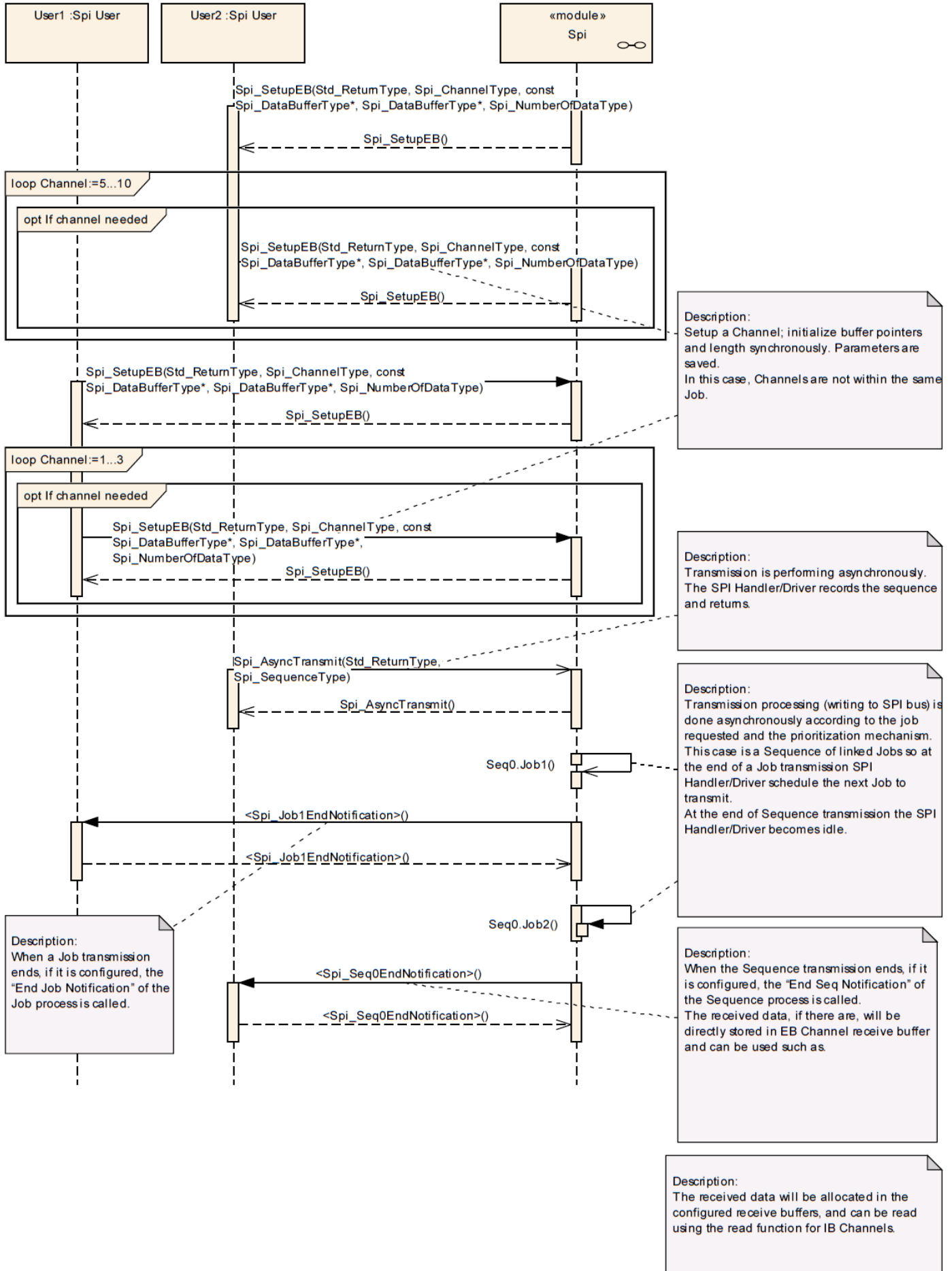


2.8.4.3 Many Channels, Many Jobs and One Sequence

The following sequence diagram shows an example of Spi_SetupEB / Spi_AsyncTransmit calls for a Sequence transmission of linked Jobs. Write or Read accesses are "User Dependent" and could be skipped when Job is just reading or writing respectively.

Example: Channels ID 0 to 3 belong to Job ID 1 (higher priority), Channels ID 4 to 10 belong to Job ID 2 (Lower priority) which has not an end notification function. These Jobs belong to the same Sequence ID 0.

Sequence	Job	Channel
ID0	ID1	ID0...ID3
	ID2	ID4...ID10



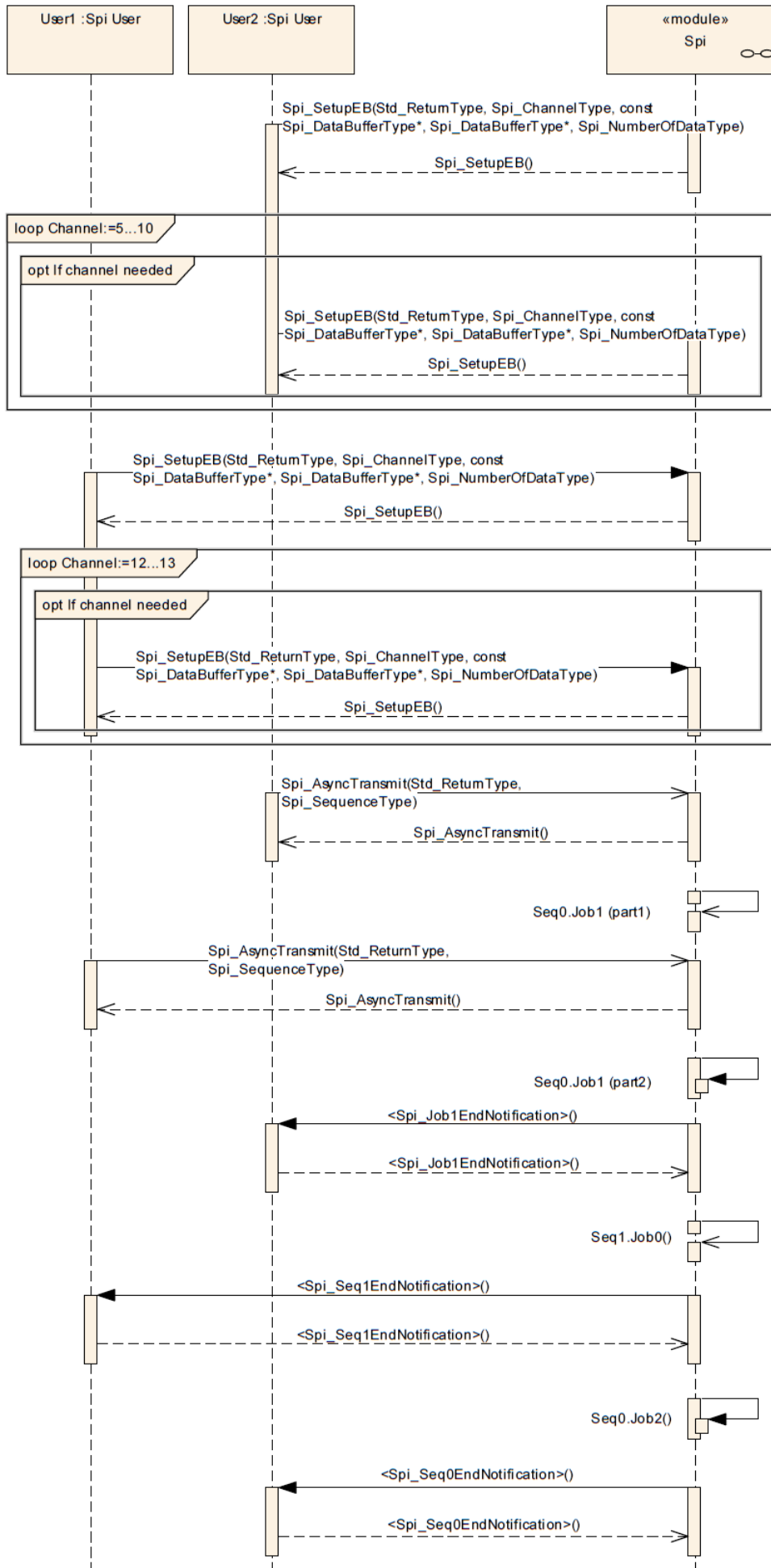
2.8.4.4 Many Channels, Many Jobs and Many Sequences

The following sequence diagram shows an example of Spi_SetupEB / Spi_AsyncTransmit calls for Sequences transmission. Write or Read accesses are "User Dependent" and could be skipped when Job is just reading or writing respectively.

Example: Channels ID 0 to 3 belong to Job ID 1 (high priority 2), Channels ID 4 to 10 belong to Job ID 2 (Low priority 1) which has not an end notification function. These Jobs belong to the same Sequence ID 0 which is configured as interruptible.

Channels ID 11 to 13 belong to Job ID 0 (higher priority 3) which belongs to Sequence ID 1 which is configured as not interruptible.

<i>Sequence</i>		<i>Job</i>		<i>Channel</i>
<i>Name</i>	<i>Interruptible</i>	<i>Name</i>	<i>Priority</i>	
ID0	Yes	ID1	2	ID0...ID3
		ID2	1	ID4...ID10
ID1	No	ID0	3	ID11...ID13



Description:
Setup a Channel; initialize buffer pointers and length synchronously. Parameters are saved.
In this case, Jobs of those Channels are not within the same Sequence.

Description:
Transmission is performing asynchronously. The SPI Handler/Driver records the sequence and returns.

Description:
Transmission processing (writing to SPI bus) is done asynchronously according to the job requested and the prioritization mechanism. This case concerns many Sequences of many Jobs so at the end of a Job transmission SPI Handler/Driver schedule the next Job to transmit.
The Job selected has the higher priority and could belong to another Sequence only if the sequence on going is configured as interruptible.
At the end of all Sequences transmission SPI Handler/Driver becomes idle.

Description:
When a Job transmission ends, if it is configured, the "End Job Notification" of the Job process is called.

Description:
When the Sequence transmission ends, if it is configured, the "End Seq Notification" of the Sequence process is called.
The received data, if there are, will be directly stored in EB Channel receive buffer and can be used such as.

2.8.4.5 Mixed Jobs Transmission

All kind of mixed Jobs transmission is possible according to the Channels configuration and the priority requirement inside Sequences.

The user knows which Channels are in use. Then, according to the types of these Channels, the appropriate methods shall be called.

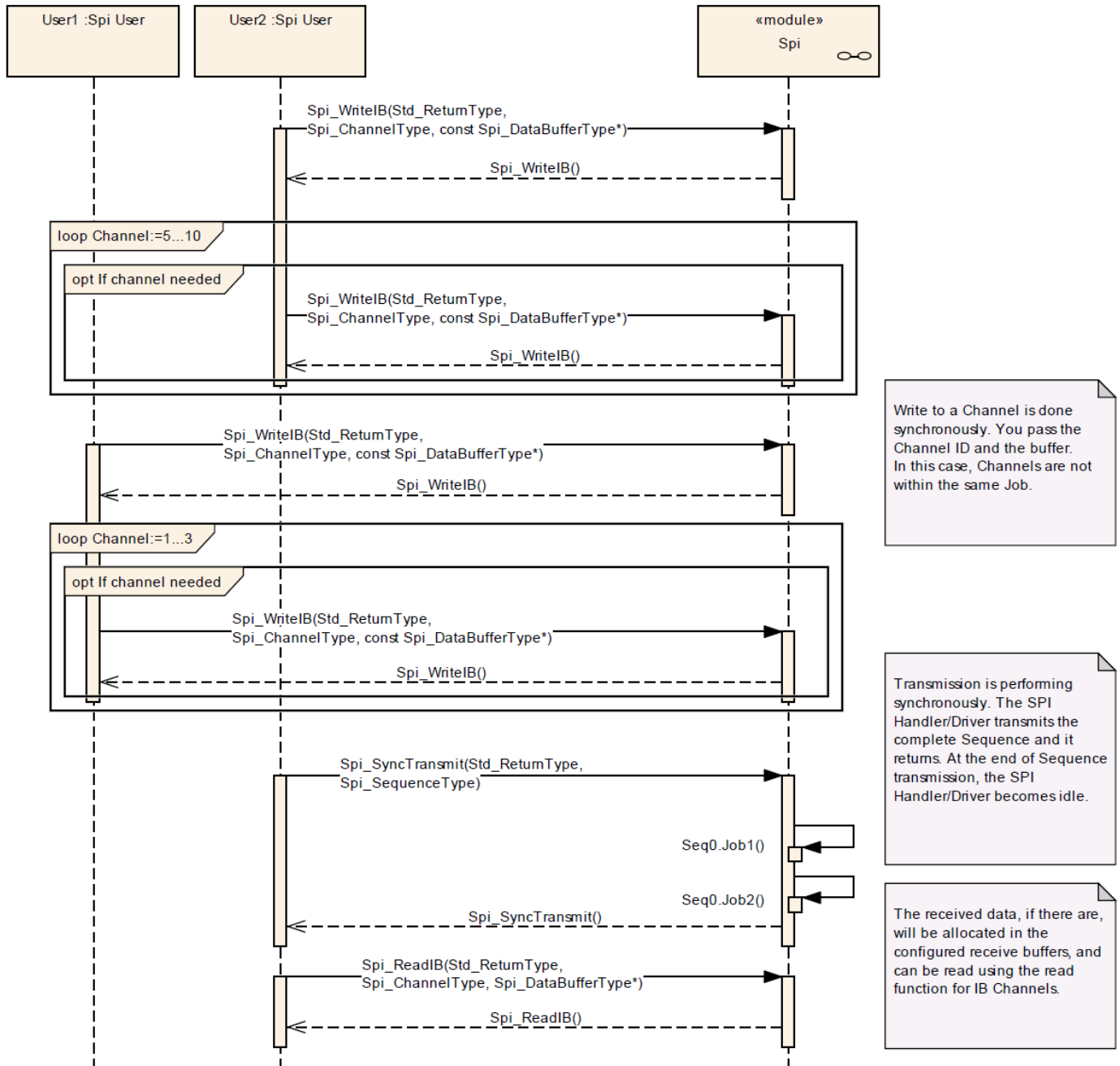
2.8.5 LEVEL 0 SyncTransmit Diagrams

2.8.5.1 Write/SyncTransmit/Read (IB): Many Channels, Many Jobs and One Sequence

The following sequence diagram shows an example of Spi_WriteIB / Spi_SyncTransmit / Spi_ReadIB calls for a Sequence transmission of linked Jobs. Write or Read steps could be skipped when Jobs are just reading or writing respectively.

Example: Channels ID 0 to 3 belong to Job ID 1 (higher priority), Channels ID 4 to 10 belong to Job ID 2 (Lower priority). These Jobs belong to the same Sequence ID 0.

Sequence	Job		Channel
	Name	Priority	
ID0	ID1	High	ID0...ID3
	ID2	Low	ID4...ID10

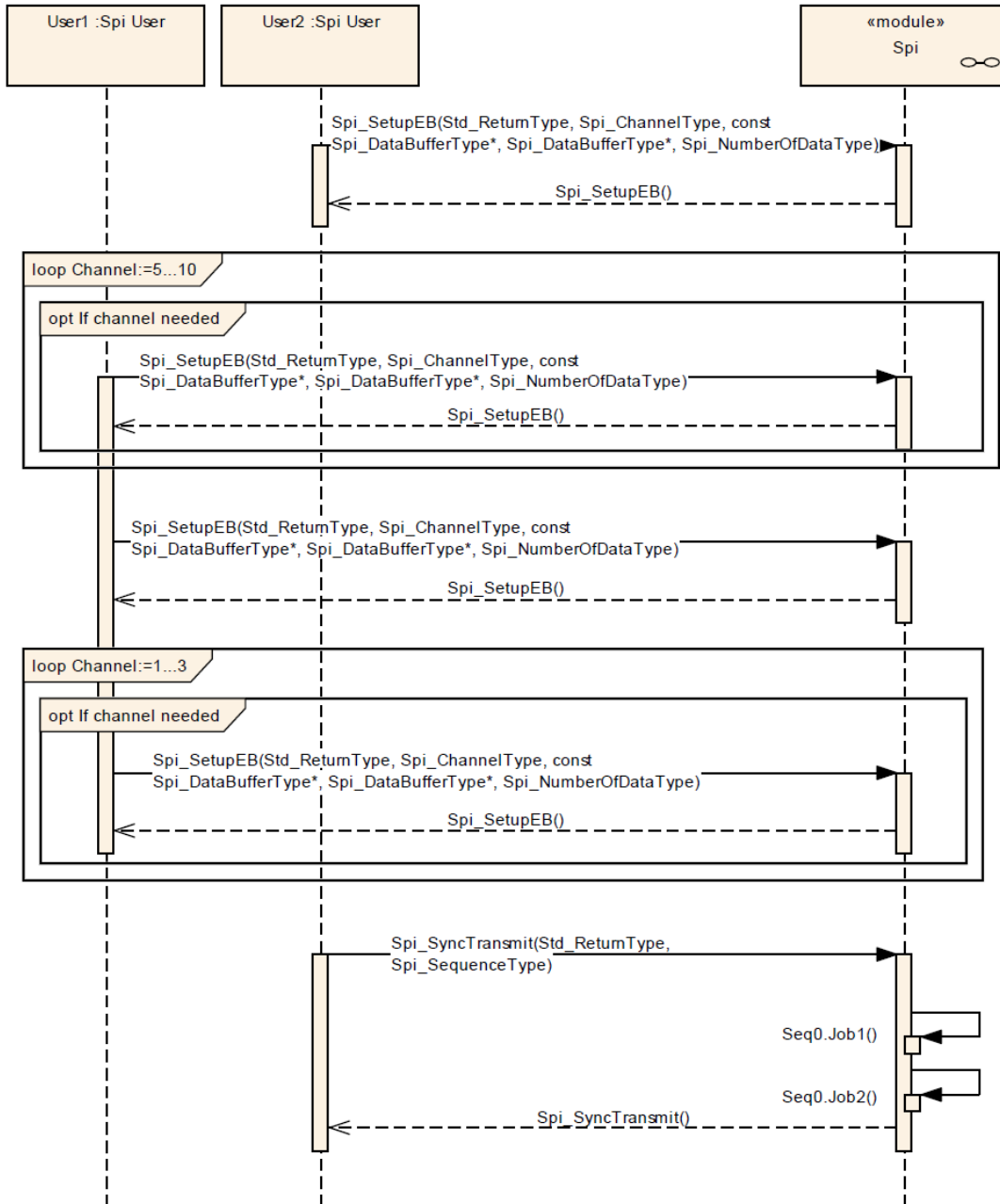


2.8.5.2 Setup/SyncTransmit (EB): Many Channels, Many Jobs and One Sequence

The following sequence diagram shows an example of Spi_SetupEB / Spi_SyncTransmit calls for a Sequence transmission of linked Jobs. Write or Read accesses are "User Dependent" and could be skipped when Job is just reading or writing respectively.

Example: Channels ID 0 to 3 belong to Job ID 1 (higher priority), Channels ID 4 to 10 belong to Job ID 2 (Lower priority). These Jobs belong to the same Sequence ID 0.

Sequence	Job	Channel
ID0	ID1	ID0...ID3
	ID2	ID4...ID10



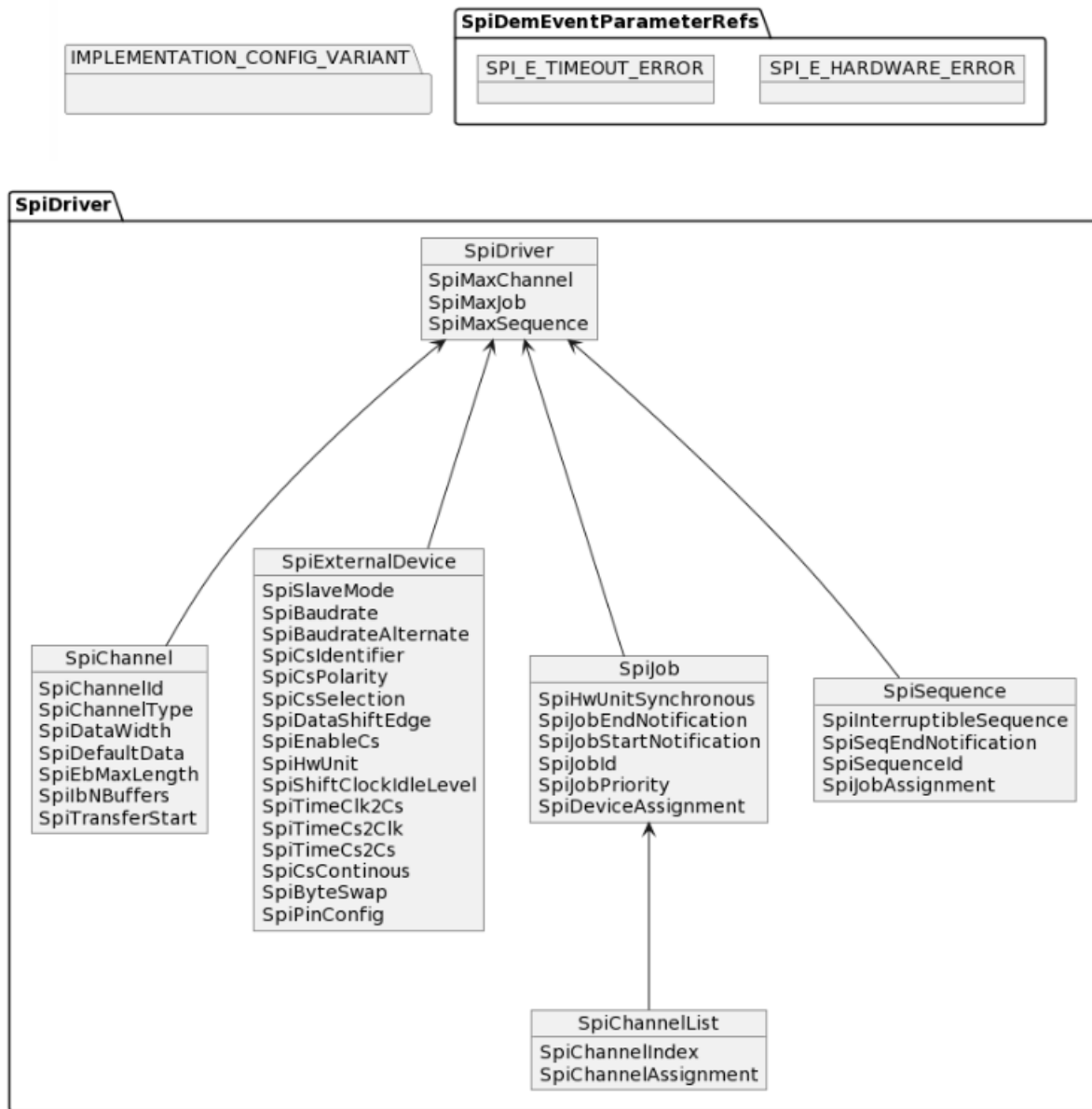
Setup a Channel; initialize buffer pointers and length synchronously. Parameters are saved. In this case, Channels are not within the same Job.

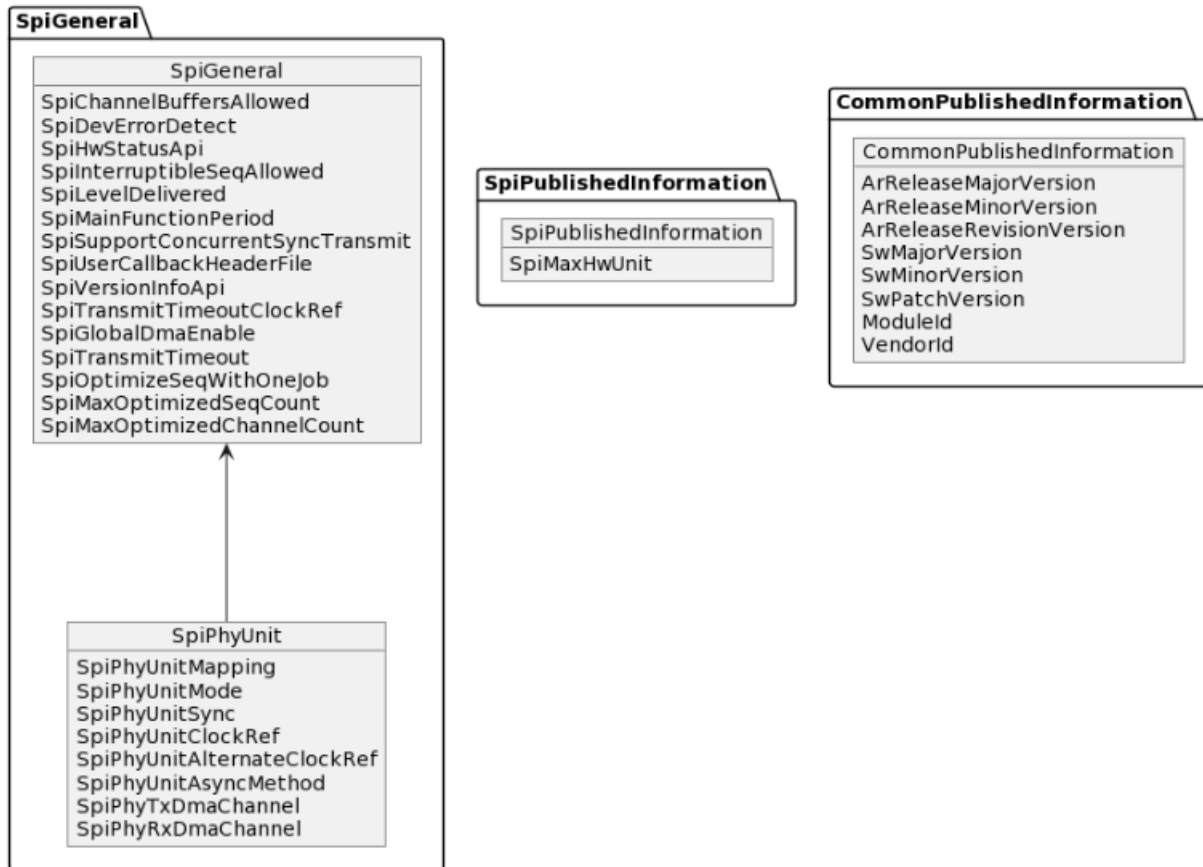
Transmission is performing synchronously. The SPI Handler/Driver transmits the complete Sequence and it returns. At the end of Sequence transmission, the SPI Handler/Driver becomes idle.

Description:
The received data, if there are, will be directly stored in EB Channel receive buffer and can be used such as.

Chapter 3 Tresos Configuration Items

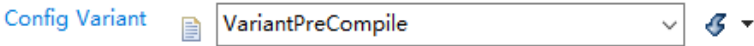
3.1 Container Inclusion Relation





3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Variable	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable: enumeration
	Label	Config Variant
	Default	VariantPreCompile
	Range	VariantLinkTime VariantPostBuild VariantPreCompile

3.2.2 SpiDriver

Container	SpiDriver	
Description	This container contains the configuration parameters and sub containers of the AUTOSAR SPI module.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1 SpiMaxChannel

Variable	SpiMaxChannel	
Description	This parameter contains the number of Channels configured. It will be gathered by tools during the configuration stage.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.2 SpiMaxJob

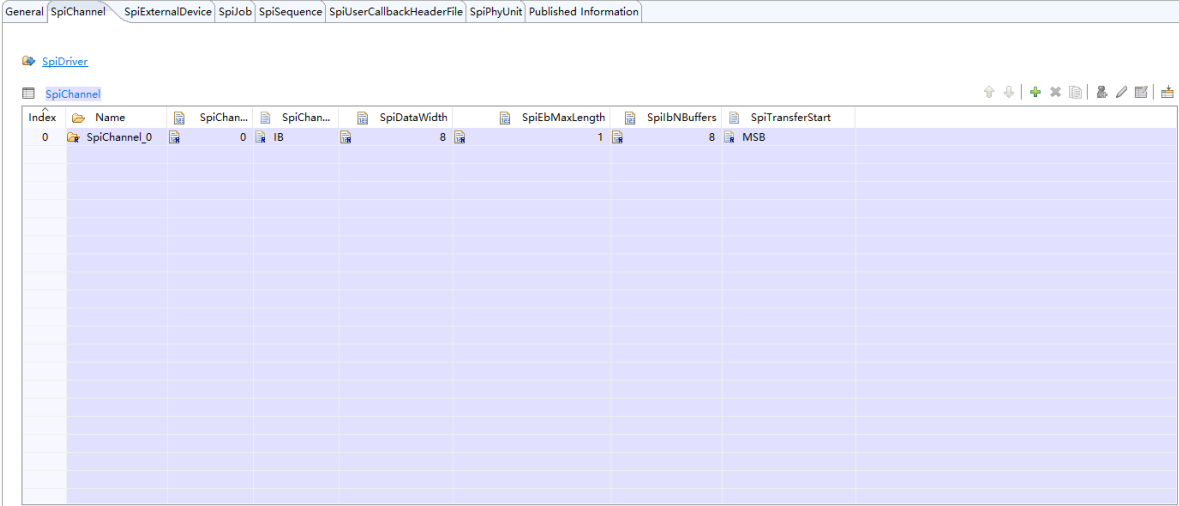
Variable	SpiMaxJob	
Description	This parameter contains the number of Jobs configured. It will be gathered by tools during the configuration stage.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.3 SpiMaxSequence

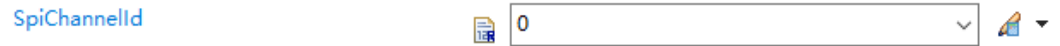
Variable	SpiMaxSequence	
Description	This parameter contains the number of Sequences configured. It will be gathered by tools during the configuration stage.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.4 SpiChannel

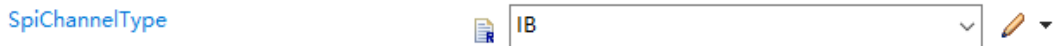
List	SpiChannel	
Description	Reference to configured DEM event to report "Timeout failure". If the reference is not configured the error shall not be reported.	

Screenshot		
Properties	Property	Value
	Type	SYMBOLIC-NAME-REFERENCE
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

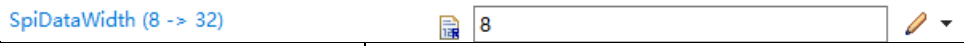
3.2.2.5 SpiChannelId

Variable	SpiChannelId	
Description	SPI Channel ID, used as parameter in SPI API functions.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.6 SpiChannelType

Variable	SpiChannelId	
Description	SPI Channel ID, used as parameter in SPI API functions.	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Range	EB IB

3.2.2.7 SpiDataWidth

Variable	SpiDataWidth	
Description	This parameter is the width of a transmitted data unit.	
Screenshot		
Properties	Property	Value
	Type	INTEGER

	Origin	AUTOSAR_ECUC
	Default	8

3.2.2.8 SpiDefaultData

Variable	SpiDefaultData	
Description	The default data to be transmitted when (for internal buffer or external buffer).	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	Default	1

3.2.2.9 SpiEbMaxLength

Variable	SpiEbMaxLength	
Description	This parameter contains the maximum size (number of data elements) of data buffers in case of EB Channels and only.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	Default	1

3.2.2.10 SpiIbNBuffers

Variable	SpiIbNBuffers	
Description	This parameter contains the maximum number of data buffers in case of IB Channels and only.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	Default	1

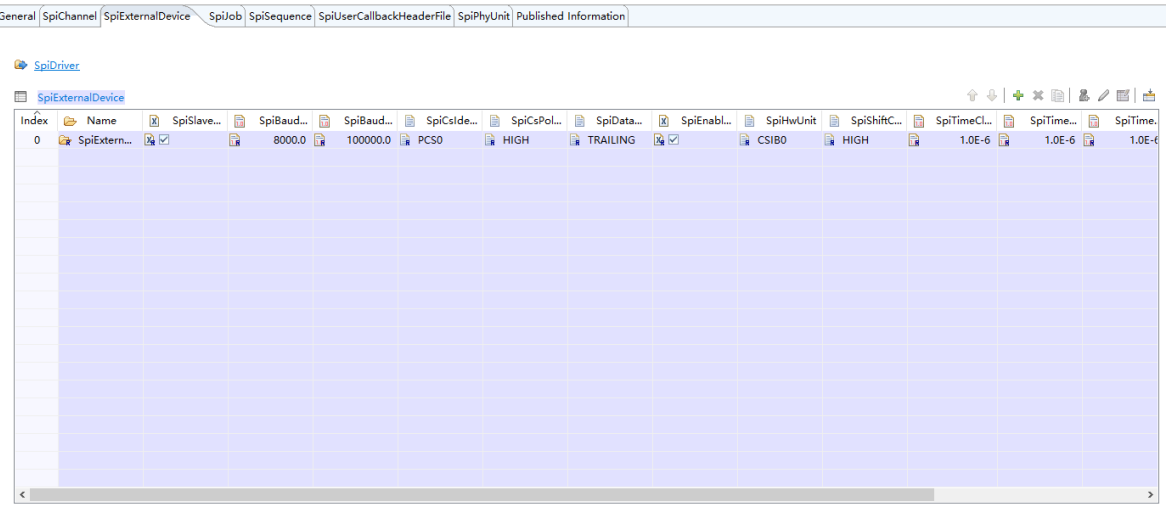
3.2.2.11 SpiTransferStart

Variable	SpiTransferStart	
Description	This parameter defines the first starting bit for transmission.	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Default	MSB
	Range	MSB LSB

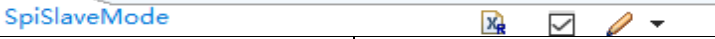
3.2.2.12 SpiChannelEcucPartitionRef

Variable	SpiChannelEcucPartitionRef	
Description	Maps an SPI Channel to zero or multiple ECUC partition to limit the access to this Channel.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	AUTOSAR_ECUC

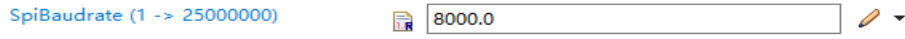
3.2.2.13 SpiExternalDevice

List	SpiExternalDevice	
Description	The communication settings of an external device. Closely linked to SpiJob.	
Screenshot		
Properties	Property	Value
	Type	MAP
	Origin	AUTOSAR_ECUC

3.2.2.14 SpiSlaveMode

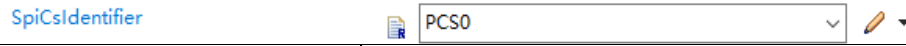
Variable	SpiSlaveMode	
Description	The communication settings of an external device. Closely linked to SpiJob.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Origin	Flagchip
	Default	false

3.2.2.15 SpiBaudrate

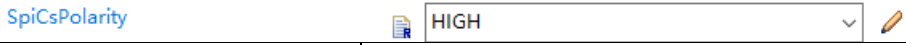
Variable	SpiBaudrate	
Description	This parameter is the communication baudrate - This parameter allows using a range of values, from the point of view of configuration tools, from Hz up to MHz.	
Screenshot		

Properties	Property	Value
	Type	FLOAT
	Origin	AUTOSAR_ECUC
	Default	100000

3.2.2.16 SpiCsIdentifier

Variable	SpiCsIdentifier	
Description	This parameter is the symbolic name to identify the Chip Select (CS) allocated to this Job.	
Screenshot		
Properties	Property	Value
	Type	STRING
	Origin	AUTOSAR_ECUC
	Default	PCS0

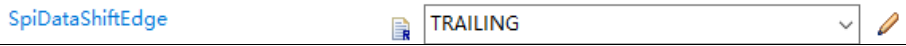
3.2.2.17 SpiCsPolarity

Variable	SpiCsPolarity	
Description	This parameter defines the active polarity of Chip Select.	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Default	HIGH
	Range	HIGH LOW

3.2.2.18 SpiCsSelection

Variable	SpiCsSelection	
Description	When the Chip select handling is enabled (see SpiEnableCs), then this parameter specifies if the chip select is handled automatically by Pe-ripheral HW engine or via general purpose IO by SPI driver.	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Default	CS_VIA_PERIPHERAL_ENGINE
	Range	CS_VIA_PERIPHERAL_ENGINE CS_VIA_GPIO

3.2.2.19 SpiDataShiftEdge

Variable	SpiDataShiftEdge	
Description	This parameter defines the SPI data shift edge.	
Screenshot		

Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Default	LEADING
	Range	LEADING TRAILING

3.2.2.20 SpiEnableCs

Variable	SpiEnableCs	
Description	This parameter enables/disables the Chip Select handling functions.	
Screenshot	   	
Properties	Property	Value
	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	true

3.2.2.21 SpiHwUnit

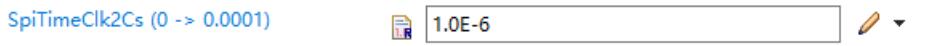
Variable	SpiHwUnit	
Description	This parameter is the symbolic name to identify the HW SPI Hardware microcontroller peripheral allocated to this Job. CSIBn references the n-th logical unit configured in SpiPhyUnit container. For example: CSIB0 references the first logical unit	
Screenshot	  CSIB0 	
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC

3.2.2.22 SpiShiftClockIdleLevel

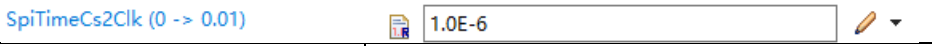
Variable	SpiShiftClockIdleLevel	
Description	This parameter defines the SPI shift clock idle level.	
Screenshot	  HIGH 	
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Default	HIGH
	Range	HIGH LOW

3.2.2.23 SpiTimeClk2Cs

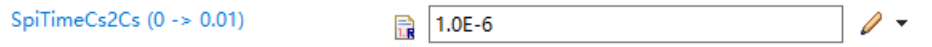
Variable	SpiTimeClk2Cs	
Description	Timing between clock and chip select in seconds (tLag : SCK to PCS Delay) – This parameter allows to use a	

	range of values from 0 up to 0.0001 Sec.	
Screenshot		
Properties	Property	Value
	Type	FLOAT
	Origin	AUTOSAR_ECUC
	Default	0.000001

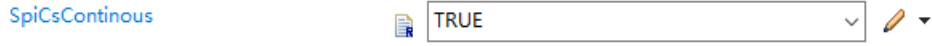
3.2.2.24 SpiTimeCs2Clk

Variable	SpiTimeCs2Clk	
Description	Timing between chip select and clock in seconds (tLead : PCS to SCK Delay) – This parameter allows to use a range of values from 0 up to 0.01 Sec.	
Screenshot		
Properties	Property	Value
	Type	FLOAT
	Origin	AUTOSAR_ECUC
	Default	0.000001

3.2.2.25 SpiTimeCs2Cs

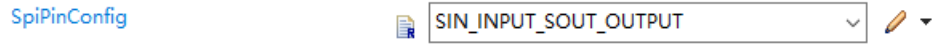
Variable	SpiTimeCs2Cs	
Description	Timing between chip select assertions in seconds (tDT : Delay Between Transfers) – This parameter allows to use a range of values from 0 up to 0.01 Sec.	
Screenshot		
Properties	Property	Value
	Type	FLOAT
	Origin	AUTOSAR_ECUC
	Default	0.000001

3.2.2.26 SpiCsContinous

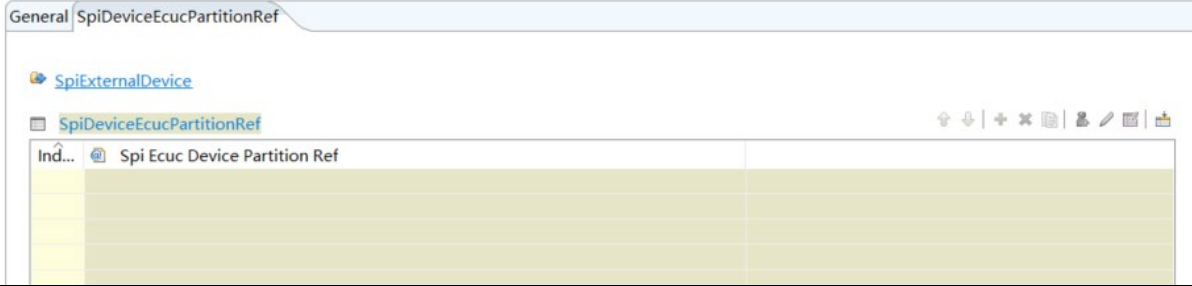
Variable	SpiCsContinous	
Description	This field determines to keep chip select asserted between frame transfers.	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Default	TRUE
	Range	TRUE FALSE

3.2.2.27 SpiPinConfig

Variable	SpiTimeCs2Clk	
Description	Timing between chip select and clock in seconds (tLead : PCS to SCK Delay) – This parameter allows to use	

	a range of values from 0 up to 0.01 Sec.	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Origin	Flagchip
	Default	SIN_INPUT_SOUT_OUTPUT
	Range	SIN_INPUT_SOUT_OUTPUT SIN_INPUT_OUTPUT SOUT_INPUT_OUTPUT SOUT_INPUT_SIN_OUTPUT

3.2.2.28 SpiDeviceEcucPartitionRef

List	SpiDeviceEcucPartitionRef	
Description	Maps an SPI external device to zero or multiple ECUC partitions to limit the access to this external device.	
Screenshot		
Properties	Property	Value
	Type	MAP
	Origin	AUTOSAR_ECUC

3.2.2.29 SpiJob

List	SpiJob	
Description	All data needed to configure one SPI-Job, amongst others the connection between the internal SPI unit and the special settings for an external device is done.	
Screenshot		
Properties	Property	Value
	Type	MAP
	MIN	1

3.2.2.30 SpiJobEndNotification

Variable	SpiJobEndNotification	
Description	This parameter is a reference to a notification function.	
Screenshot		
Properties	Property	Value
	Type	FUNCTION-NAME
	Origin	AUTOSAR_ECUC
	Default	NULL_PTR

3.2.2.31 SpiJobStartNotification

Variable	SpiJobStartNotification	
Description	This parameter is a reference to a notification function.	
Screenshot		
Properties	Property	Value
	Type	FUNCTION-NAME
	Origin	AUTOSAR_ECUC
	Default	NULL_PTR

3.2.2.32 SpiJobId

Variable	SpiJobId	
Description	SPI Job ID, used as parameter in SPI API functions.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC

3.2.2.33 SpiJobPriority

Variable	SpiJobPriority	
Description	Priority set accordingly to SPI093: 0, lowest, 3, highest priority	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	Default	0

3.2.2.34 SpiDeviceAssignment

Variable	SpiDeviceAssignment	
Description	Reference to the external device used by this job.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	AUTOSAR_ECUC
	Default	SIN_INPUT_SOUT_OUTPUT

3.2.2.35 SpiChannelList

List	SpiChannelList
Description	References to SPI channels and their order within the Job.

Screenshot		
Properties	Property	Value
	Type	MAP
	Min	1

3.2.2.36 SpiChannelIndex

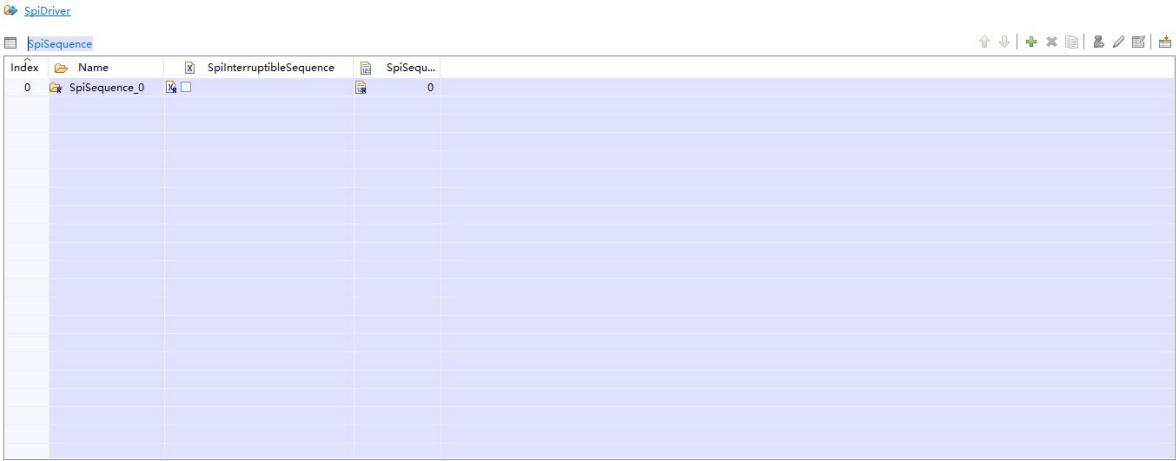
Variable	SpiChannelIndex	
Description	This parameter specifies the order of Channels within the Job.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC

3.2.2.37 SpiChannelAssignment

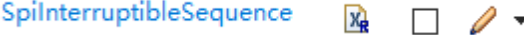
Variable	SpiChannelAssignment	
Description	A job references several channels.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	AUTOSAR_ECUC

3.2.2.38 SpiSequence

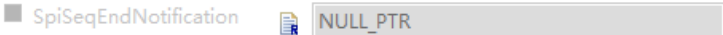
List	SpiSequence	
Description	Reference to the external device used by this job.	

Screenshot		
Properties	Property	Value
	Type	MAP
	Min	1

3.2.2.39 SpiInterruptibleSequence

Variable	SpiInterruptibleSequence	
Description	This parameter allows or not this Sequence to be suspended by another one.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	AUTOSAR_ECUC
	Default	False

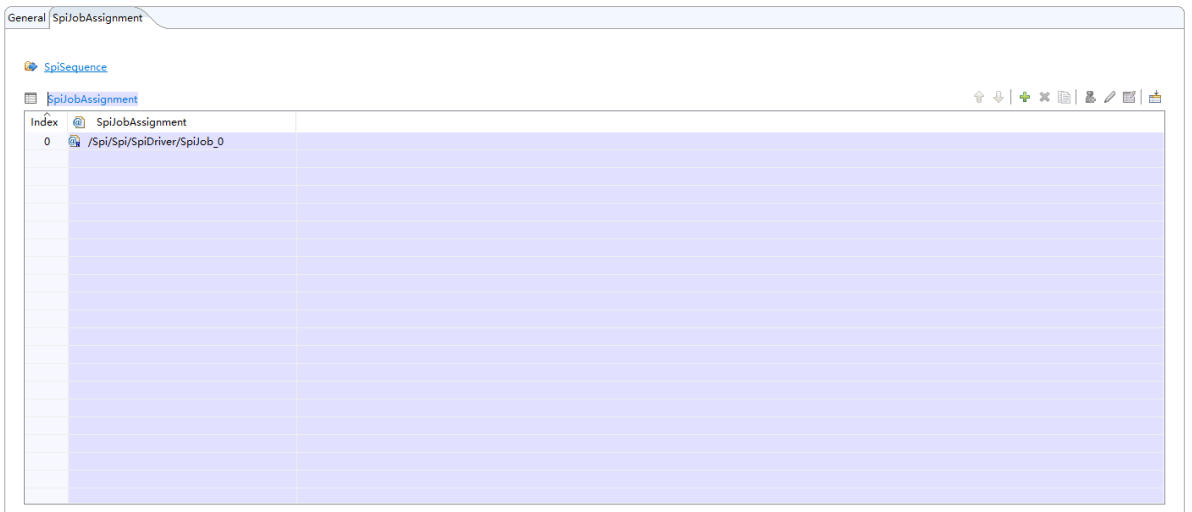
3.2.2.40 SpiSeqEndNotification

Variable	SpiSeqEndNotification	
Description	This parameter is a reference to a notification function.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	AUTOSAR_ECUC
	Default	NULL_PTR

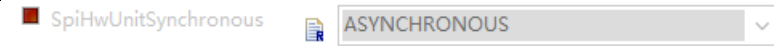
3.2.2.41 SpiSequenceld

Variable	SpiSequenceld	
Description	Sequence ID of configured SPI Sequence.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC

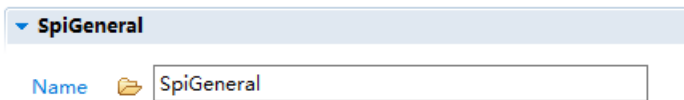
3.2.2.42 SpiJobAssignment

List	SpiJobAssignment	
Description	Reference to the external device used by this job.	
Screenshot		
Properties	Property	Value
	Type	MAP
	Min	1

3.2.2.43 SpiHwUnitSynchronous

Variable	SpiHwUnitSynchronous	
Description	If SpiHwUnitSynchronous is set to SYNCHRONOUS, the SpiJob uses its containing SpiDriver in a synchronous manner.	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Origin	AUTOSAR_ECUC
	Default	ASYNCHRONOUS
	Range	ASYNCHRONOUS SYNCHRONOUS

3.2.3 SpiGeneral

Container	SpiGeneral	
Description	General configuration settings for SPI-Handler.	
Screenshot		

3.2.3.1 SpiMulticoreSupport

Container	SpiMulticoreSupport	
Description	When this parameter is enabled, multi-core feature will be used in SPI driver. That means mapping the SPI driver to multiple ECUC partitions to make the module API available in this partition.	
Screenshot		

Properties	Property	Value
	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	FALSE

3.2.3.2 SpiKernelEcucPartitionRef

Container	SpiKernelEcucPartitionRef	
Description	Maps the SPI kernel to zero or one ECUC partition to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the SPI driver is mapped to. Since the driver implementation only supports multi-core type II, this option is not supported.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC
	Default	NONE

3.2.3.3 SpiLevelDelivered

Variable	SpiLevelDelivered	
Description	Selects the SPI Handler/Driver level of scalable functionality that is available and delivered.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	AUTOSAR_ECUC
	Default	2

3.2.3.4 SpiChannelBuffersAllowed

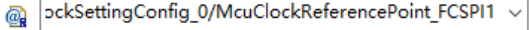
Variable	SpiChannelBuffersAllowed	
Description	Selects the SPI Handler/Driver Channel Buffers usage allowed and delivered.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Default	0
	Editable	false

3.2.3.5 SpiMainFunctionPeriod

Variable	SpiMainFunctionPeriod	
Description	This parameter defines the cycle time of the function Spi_MainFunction_Handling in seconds.	
Screenshot		
Properties	Property	Value

	Type	FLOAT
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Default	0.01

3.2.3.6 SpiTransmitTimeoutClockRef

Variable	SpiTransmitTimeoutClockRef	
Description	Reference to the CPU clock source configuration, which is set into the MCU driver configuration.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	AUTOSAR_ECUC
	REF	ASPathDataOfSchema:/AUTOSAR/EcucDefs/Mcu/McuModuleConfigurati on/McuClockSettingConfig/McuClockReferencePoint

3.2.3.7 SpiTransmitTimeout

Variable	SpiTransmitTimeout	
Description	Timeout value (microseconds) used to wait for TX/RX transmission to complete one frame.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Origin	Flagchip
	SymbolicNameValue	false
	Default	50000

3.2.3.8 SpiGlobalDmaEnable

Variable	SpiGlobalDmaEnable	
Description	For each SPI unit a transferring method can be configured: FIFO or DMA.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Origin	Flagchip
	SymbolicNameValue	false
	Default	false

3.2.3.9 SpiInterruptibleSeqAllowed

Variable	SpiInterruptibleSeqAllowed	
Description	Switches the Interruptible Sequences handling functionality ON or OFF. This parameter depends on SPI_LEVEL_DELIVERED value.	
Screenshot		
Properties	Property	Value

	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	false

3.2.3.10 SpiSupportConcurrentSyncTransmit

Variable	SpiSupportConcurrentSyncTransmit	
Description	Specifies whether concurrent Spi_SyncTransmit() calls for different sequences shall be configurable.	
Screenshot	SpiSupportConcurrentSyncTransmit  ☐ 	
Properties	Property	Value
	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Default	false

3.2.3.11 SpiDevErrorDetect

Variable	SpiDevErrorDetect	
Description	Switches the Development Error Detection and Notification ON or OFF.	
Screenshot	SpiDevErrorDetect  ☒ 	
Properties	Property	Value
	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	false

3.2.3.12 SpiHwStatusApi

Variable	SpiHwStatusApi	
Description	Switches the Spi_GetHWUnitStatus function ON or OFF.	
Screenshot	SpiHwStatusApi  ☒ 	
Properties	Property	Value
	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	true

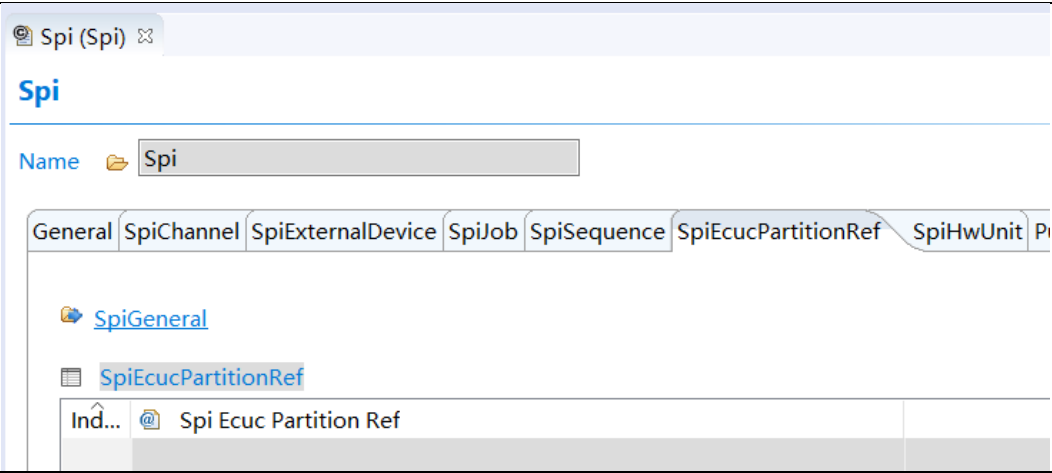
3.2.3.13 SpiCancelApi

Container	SpiCancelApi	
Description	Switches the Spi_Cancel function ON or OFF.	
Screenshot	SpiCancelApi  ☒ 	
Properties	Property	Value
	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	TRUE

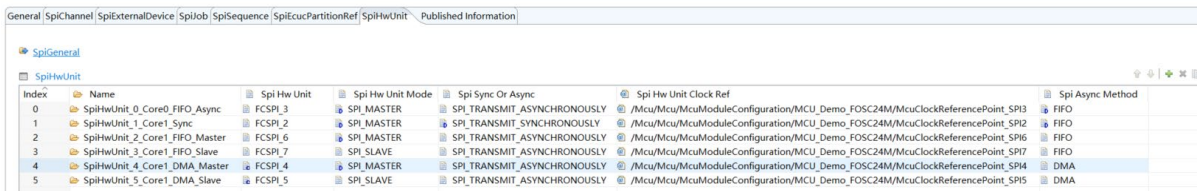
3.2.3.14 SpiVersionInfoApi

Variable	SpiVersionInfoApi	
Description	Switches the Spi_GetVersionInfo function ON or OFF.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	false

3.2.3.15 SpiDeviceEcucPartitionRef

List	SpiEcucPartitionRef	
Description	Maps the SPI driver to zero or multiple ECUC partitions to make the driver API available in the according partition.	
Screenshot		
Properties	Property	Value
	Type	MAP
	Origin	AUTOSAR_ECUC

3.2.3.16 SpiHwUnit

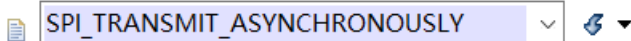
List	SpiHwUnit	
Description	Logical to Physical SPI Bus mapping.	
Screenshot		
Properties	Property	Value
	Type	MAP

3.2.3.17 SpiHwUnitMode

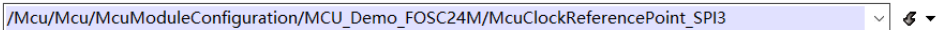
Variable	SpiHwUnitMode	
Description	Select between SPI_MASTER and SPI_SLAVE modes.	
Screenshot		

Properties	Property	Value
	Type	ENUMERATION
	Origin	Flagchip
	SymbolicNameValue	false
	Default	SPI_MASTER
	Range	SPI_MASTER SPI_SLAVE

3.2.3.18 SpiSyncOrAsync

Variable	SpiSyncOrAsync	
Description	Specifies if this HwUnit can only do sync transfers or Async transfers .	
Screenshot	 	
Properties	Property	Value
	Type	ENUMERATION
	Origin	Flagchip
	SymbolicNameValue	false
	Default	SPI_TRANSMIT_SYNCHRONOUSLY
	Range	SPI_TRANSMIT_SYNCHRONOUSLY SPI_TRANSMIT_ASYNCHRONOUSLY

3.2.3.19 SpiHwUnitClockRef

Variable	SpiHwUnitClockRef	
Description	Reference to the SPI clock source configuration, which is set into the MCU driver configuration.	
Screenshot	 	
Properties	Property	Value
	Type	REFERENCE
	Origin	Flagchip
	SymbolicNameValue	false
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/Mcu/McuModuleConfigurati on/McuClockSettingConfig/McuClockReferencePoint

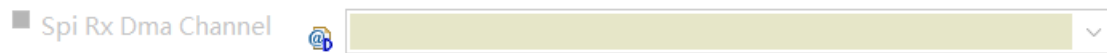
3.2.3.20 SpiAsyncMethod

Variable	SpiAsyncMethod	
Description	Transfer method of the periperal: DMA, FIFO.	
Screenshot	 	
Properties	Property	Value
	Type	ENUMERATION
	Origin	Flagchip
	SymbolicNameValue	false
	Default	FIFO

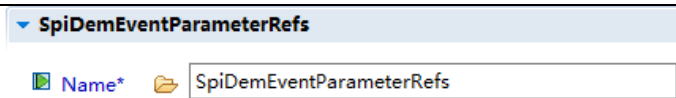
3.2.3.21 SpiTxDmaChannel

Variable	SpiTxDmaChannel	
Description	Select logical DMA channel for Spi Hw unit to transive data.	
Screenshot		
Properties	Property	Value
	Type	CHOICE-REFERENCE
	Origin	Flagchip
	Ref	ASPathDataOfSchema:/FC7300FXXX/Mcl/MclConfigSet/DMAChannel

3.2.3.22 SpiRxDmaChannel

Variable	SpiRxDmaChannel	
Description	Select logical DMA channel for Spi Hw unit to receive data.	
Screenshot		
Properties	Property	Value
	Type	CHOICE-REFERENCE
	Origin	Flagchip
	Ref	ASPathDataOfSchema:/FC7300FXXX/Mcl/MclConfigSet/DMAChannel

3.2.4 SpiDemEventParameterRefs

Container	SpiDemEventParameterRefs	
Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus API in case the corresponding error occurs.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.4.1 SPI_E_HARDWARE_ERROR

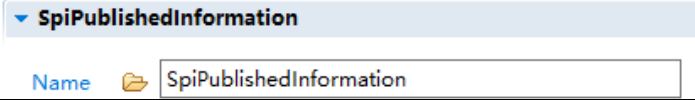
Variable	SPI_E_HARDWARE_ERROR	
Description	Reference to configured DEM event to report "Hardware failure". If the reference is not configured the error shall not be reported.	
Screenshot		
Properties	Property	Value
	Type	SYMBOLIC-NAME-REFERENCE
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.4.2 SPI_E_TIMEOUT_ERROR

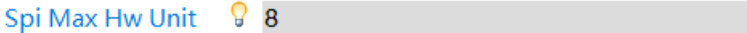
Variable	SPI_E_TIMEOUT_ERROR	
Description	Reference to configured DEM event to report "Timeout failure". If the reference is not configured the error shall not be reported.	

Screenshot		
Properties	Property	Value
	Type	SYMBOLIC-NAME-REFERENCE
	Origin	Flagchip
	SymbolicNameValue	false

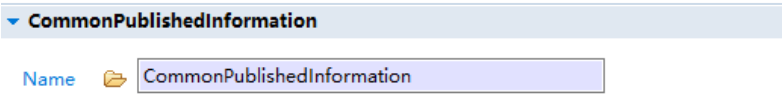
3.2.5 SpiPublishedInformation

Container	SpiPublishedInformation	
Description	Container holding all SPI specific published information parameters.	
Screenshot		
Properties	N/A	

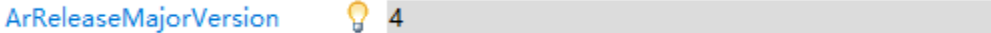
3.2.5.1 SpiMaxHwUnit

Variable	SpiMaxHwUnit	
Description	Number of different SPI hardware microcontroller peripherals (units/busses) available and handled by this SPI Handler/Driver module. We calculate the count of SpiGeneral/SpiPhyUnit/* instead.	
Screenshot		
Properties	Property	Value
	Type	IDENTIFIABLE
	Origin	AUTOSAR_ECUC
	Default	8

3.2.6 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		

3.2.6.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	4

3.2.6.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	6

3.2.6.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.6.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.6.5 SwMinorVersion

Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	3

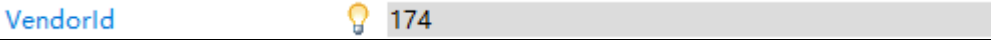
3.2.6.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.6.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	83

3.2.6.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	174

3.2.7 SpiExtendedFeatures

3.2.7.1 SpiAllowBigSizeCollections

Variable	SpiAllowBigSizeCollections	
Description	A feature to allow more than 256 sequences, jobs, and channels	
Screenshot		
Properties	Property	Value
	Type	Variable : Boolean
	Origin	Flagchip
	SymbolicNameValue	false
	Default	false

3.2.7.2 SpiEnableHWUnitAsyncMode

Variable	SpiEnableHWUnitAsyncMode	
Description	If SpiLevelDelivered=2 and SpiEnableHWUnitAsyncMode is enabled,function "Spi_SetHWUnitAsyncMode()" will be enabled ,which can set each HWUnitto asynchronous mode (POLLING or INTERRUPT) independently	
Screenshot	Spi Enable Hw Unit AsyncMode    	
Properties	Property	Value
	Type	Variable : Boolean
	Origin	Flagchip
	SymbolicNameValue	false
	Default	false

3.2.7.3 SpiJobStartNotificationEnable

Variable	SpiJobStartNotificationEnable	
Description	It's a setting only for SpiJobStartNotification configuration.	
Screenshot	Spi Job Start Notification Enable    	
Properties	Property	Value
	Type	Variable : Boolean
	Origin	Flagchip
	SymbolicNameValue	false
	Default	false

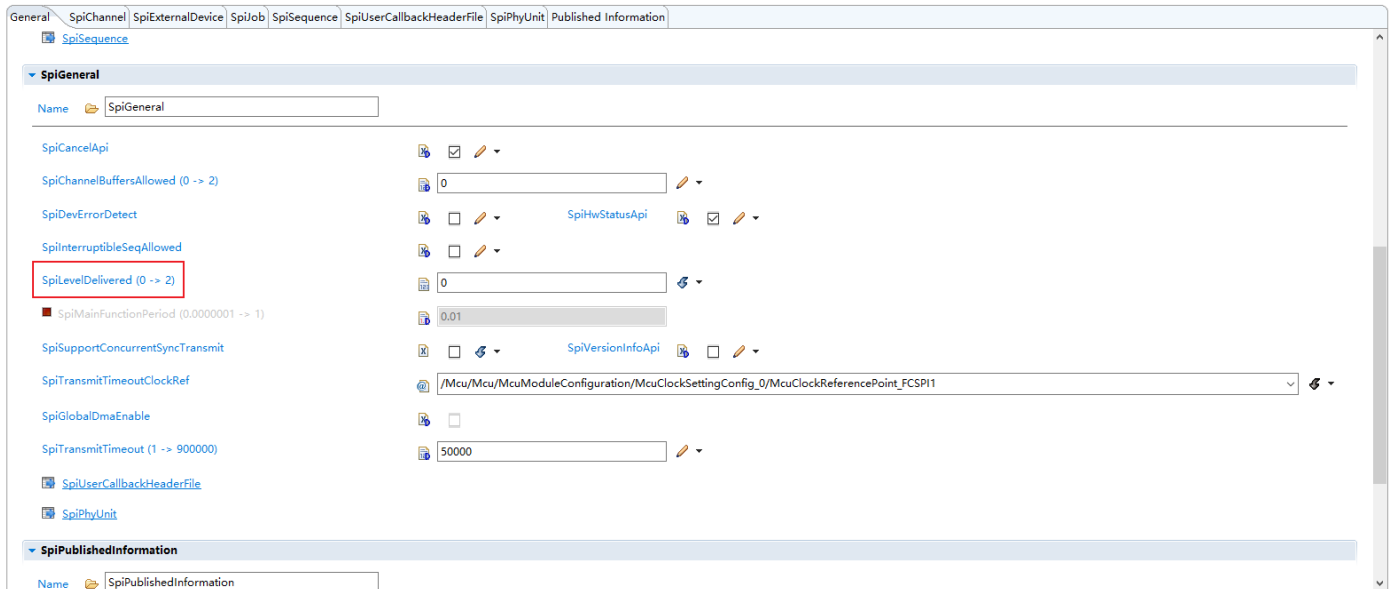
3.2.7.4 SpiDisableDemReportErrorStatus

Variable	SpiDisableDemReportErrorStatus	
Description	Switches the Diagnostic Error Reporting and Notification OFF	
Screenshot	Spi Disable Production Error Reporting    	
Properties	Property	Value
	Type	Variable : Boolean
	Origin	Flagchip
	SymbolicNameValue	false
	Default	false

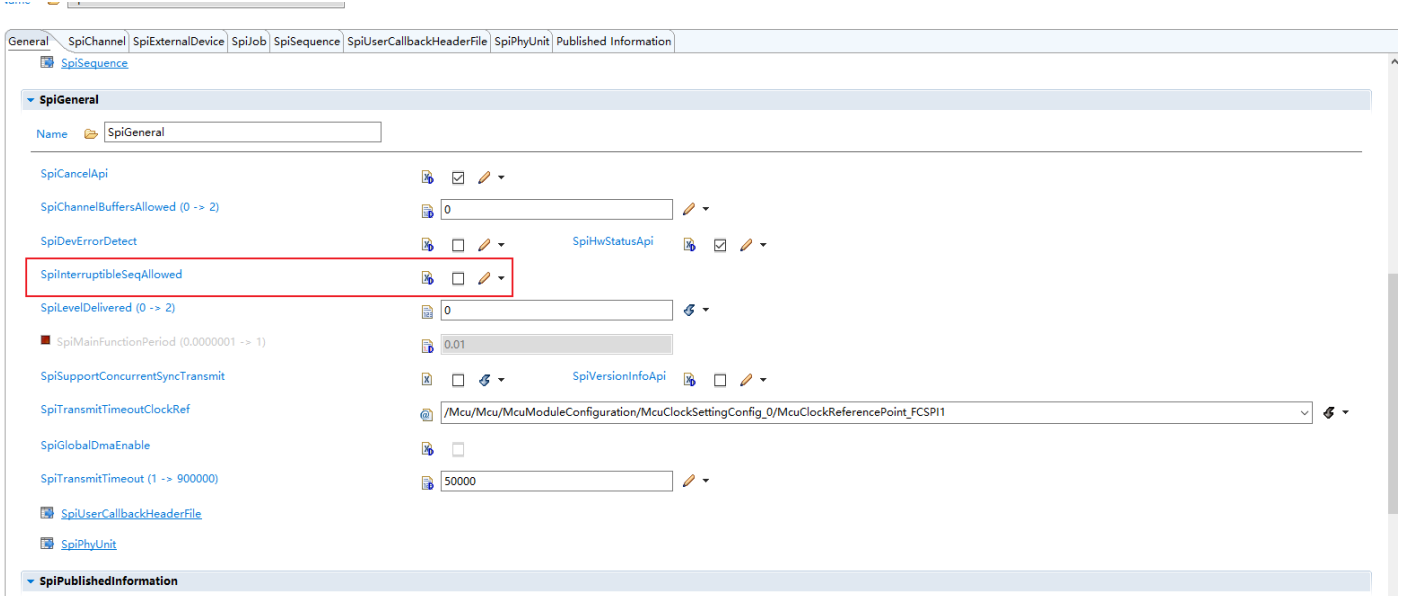
Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

- 1) According to AUTOSAR SPI Specification, Notification is only valid in asynchronous mode of LEVEL 1 and LEVEL 2. Therefore, please make sure that "SpiLevelDelivered" is set to 1 / 2 and Notification pointer must not be NULL_PTR.



- 2) If the interruptible sequence is needed, you should enable the item "SpiInterruptibleSeqAllowed" in the tab "General" first, then enable the item "SpiInterruptibleSequence" for the specific sequence.



SpiSequence

Name  SpiSequence_0

General SpiJobAssignment

SpiInterruptibleSequence    

 SpiSeqEndNotification  NULL_PTR

SpiSequenceId  0  

 [SpiJobAssignment](#)

4.2 SPI Usage Common Steps

Basically, the SPI module could be configured by following the below steps:

- 1) Configure SPI channels in the tab "SpiChannel".

 [SpiChannel](#)

Index	Name	SpiChannelId	SpiChannelType	SpiDataWidth	SpiEbMaxLength	SpiNbBuffers	SpiTrans...
0	 SpiChannel_0	 0	 IB	 8	 1	 4	 MSB
1	 SpiChannel_1	 1	 IB	 8	 1	 2	 MSB

- 2) Configure SPI Jobs in the tab "SpiJob" and assign channels configured in Step 1 to adequate Jobs.

 [SpiDriver](#)

 [SpiJob](#)

Index	Name	SpiJobId	SpiJobPriority	SpiDeviceAssignment
0	 SpiJob_0	 0	 0	 @ /Spi/Spi/SpiDriver/SpiExternalDevice_0

General SpiChannelList

 [SpiJob](#)

 [SpiChannelList](#)

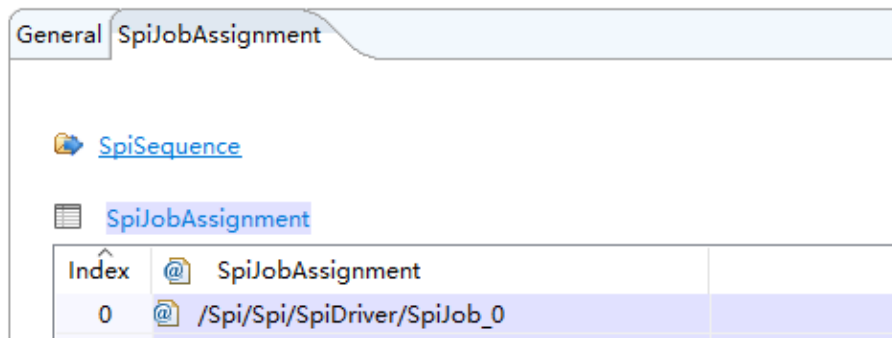
Index	Name	SpiChannelIndex	SpiChannelAssignment
0	 SpiChannelList_0	 0	 @ /Spi/Spi/SpiDriver/SpiChannel_0
1	 SpiChannelList_1	 1	 @ /Spi/Spi/SpiDriver/SpiChannel_1

- 3) Configure SPI Sequences in the tab "SpiSequence" and assign Jobs configured in Step 2 to adequate Sequences.

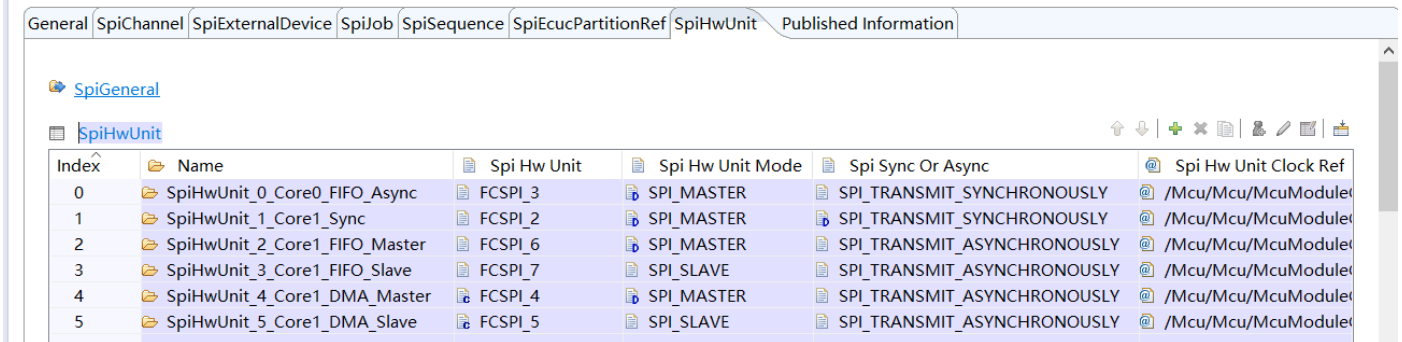
 [SpiDriver](#)

 [SpiSequence](#)

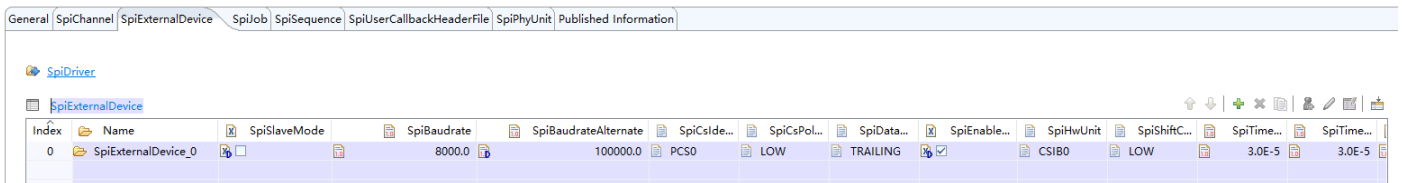
Index	Name	SpiInterruptibleSequence	SpiSequenceId
0	 SpiSequence_0	 	 0



4) Configure the FCSPI Hardware unit used in the tab "SpiHwUnit".



5) Configure external devices in the tab "SpiExternalDevice" according to the features of devices communicating with SPI buses.



6) Enable necessary configuration items in the tab "General".

Spi

Name

General SpiChannel SpiExternalDevice SpiJob SpiSequence SpiUserCallbackHeaderFile SpiPhyUnit Published Information

SpiGeneral

Name

SpiCancelApi ☒

SpiChannelBuffersAllowed (0 -> 2)

SpiDevErrorDetect ☐ SpiHwStatusApi ☒

SpiInterruptibleSeqAllowed ☐

SpiLevelDelivered (0 -> 2)

SpiMainFunctionPeriod (0.0000001 -> 1)

SpiSupportConcurrentSyncTransmit ☐ SpiVersionInfoApi ☐

SpiTransmitTimeoutClockRef

SpiGlobalDmaEnable ☐

SpiTransmitTimeout (1 -> 900000)

[SpiUserCallbackHeaderFile](#)

[SpiPhyUnit](#)

SpiPublishedInformation

Name

7) Generate configuration files.